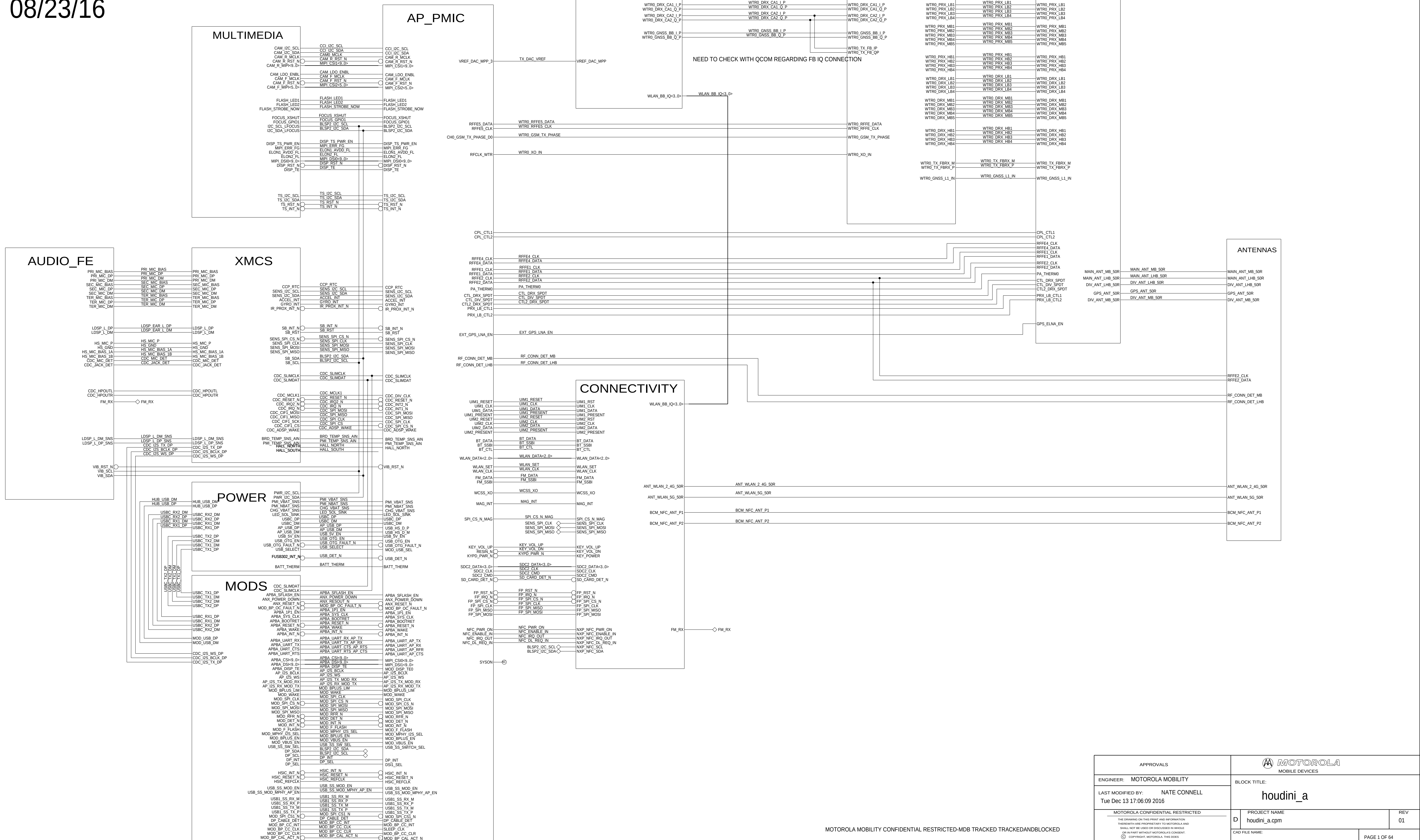


HOUDINI
SYSTEM 17
08/23/16



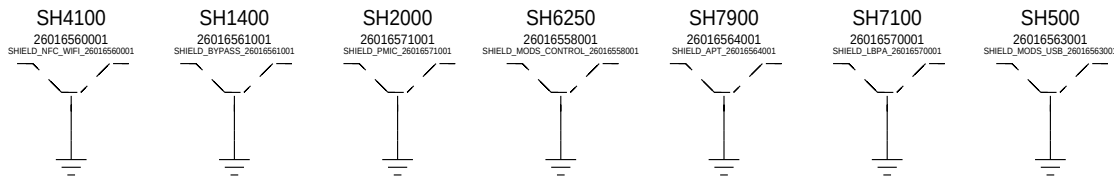
APPROVALS	 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA MOBILITY	BLOCK TITLE:		
LAST MODIFIED BY: NATE CONNELL Tue Dec 13 17:06:09 2016	houdini_a		
MOTOROLA CONFIDENTIAL RESTRICTED	D	PROJECT NAME houdini_a.cpm	REV 01
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HOUDINI ME/SHIELDS

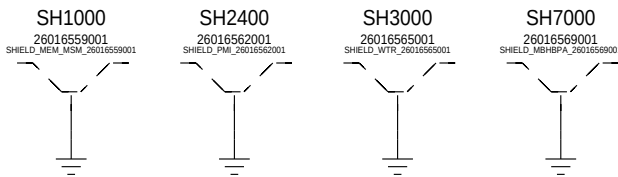
REFERENCE DESIGNATOR RANGES BY BLOCK

- 500 - 599: USER INPUT
- 600 - 799: POWER
- 1000-2999: MSM8953 + PM8953
- 3000-3199: WTR3925
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO FE
- 5000-5499: DISPLAY
- 5500-5999: CAMERA
- 6000-6999: XMCS
- 7000-7999: RFFE
- 8000 - 8099: ANTENNA

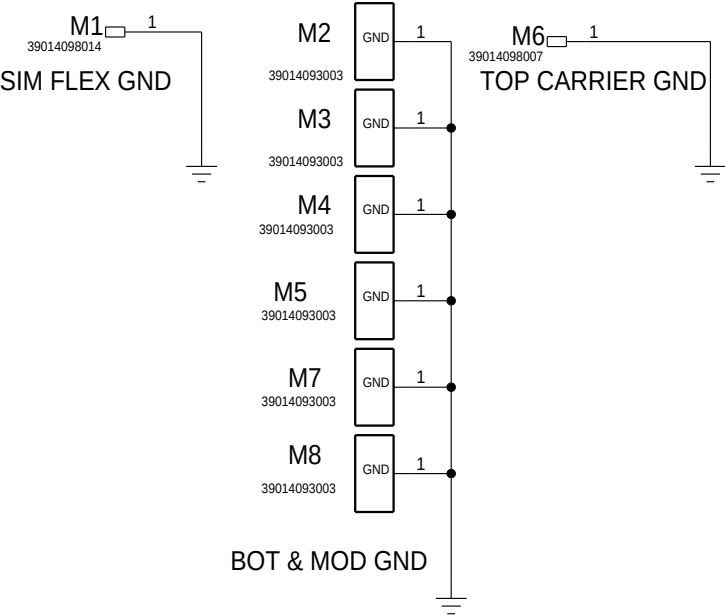
TOP SIDE PCB SHIELD



BOTTOM SIDE PCB SHIELD



GROUND CONTACTS

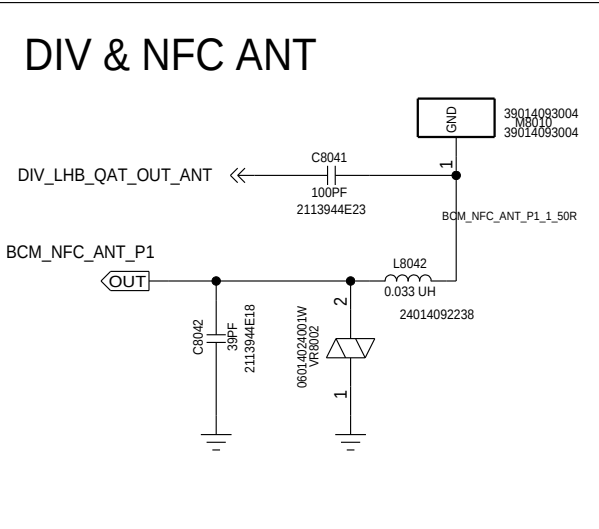
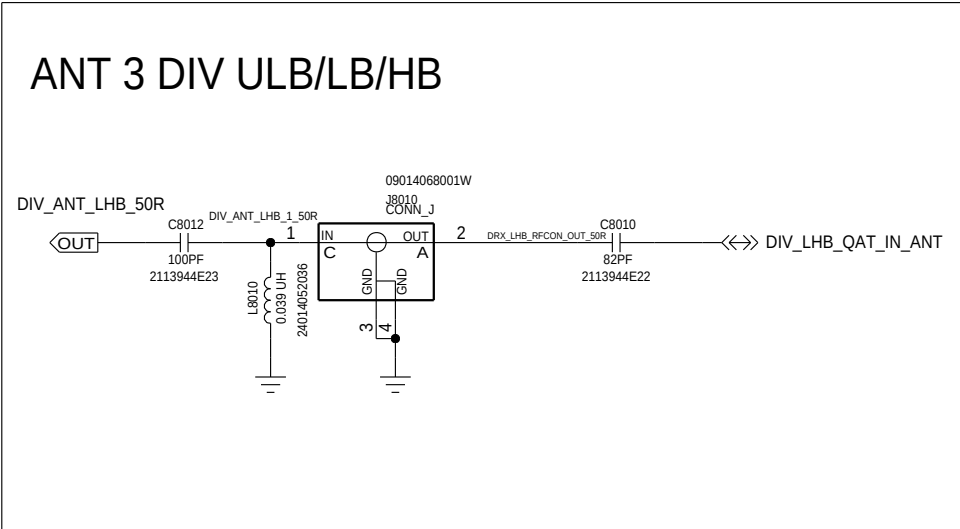
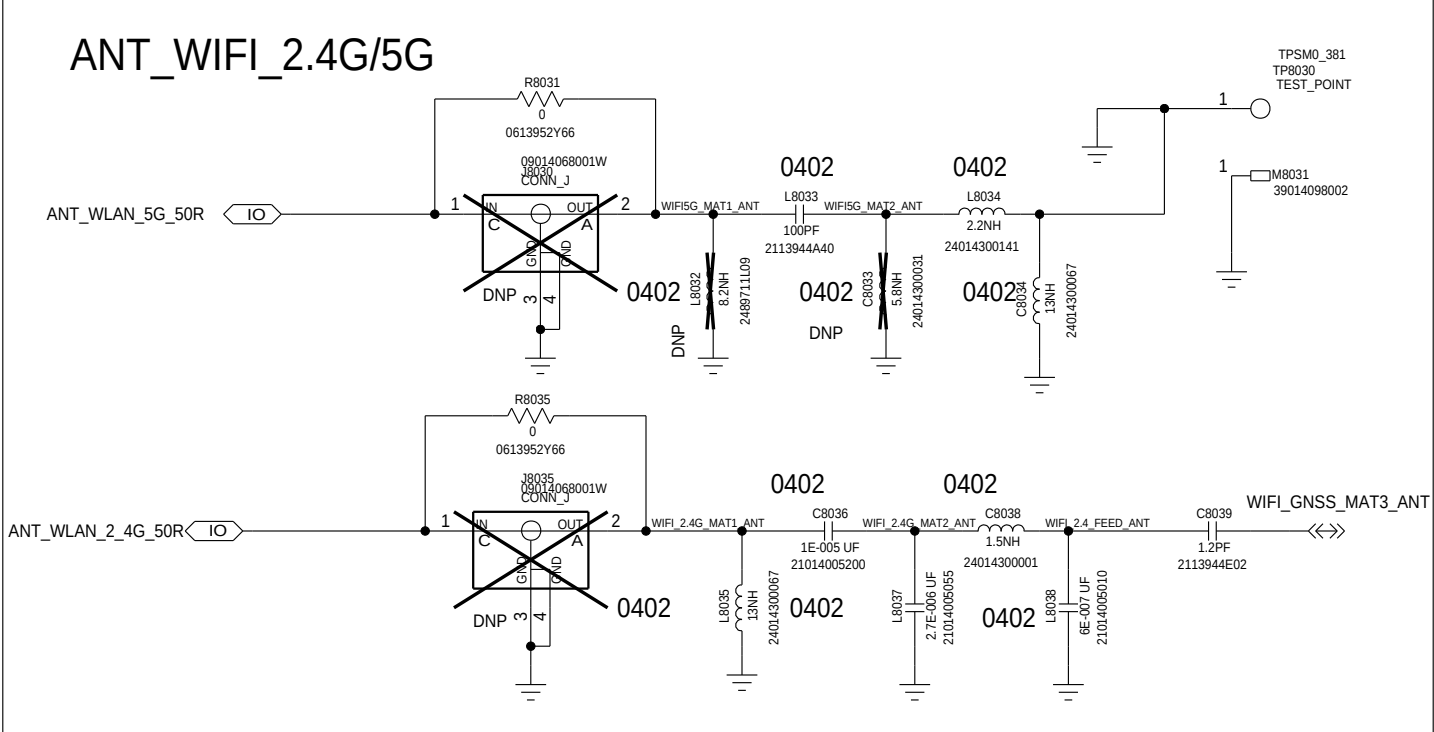
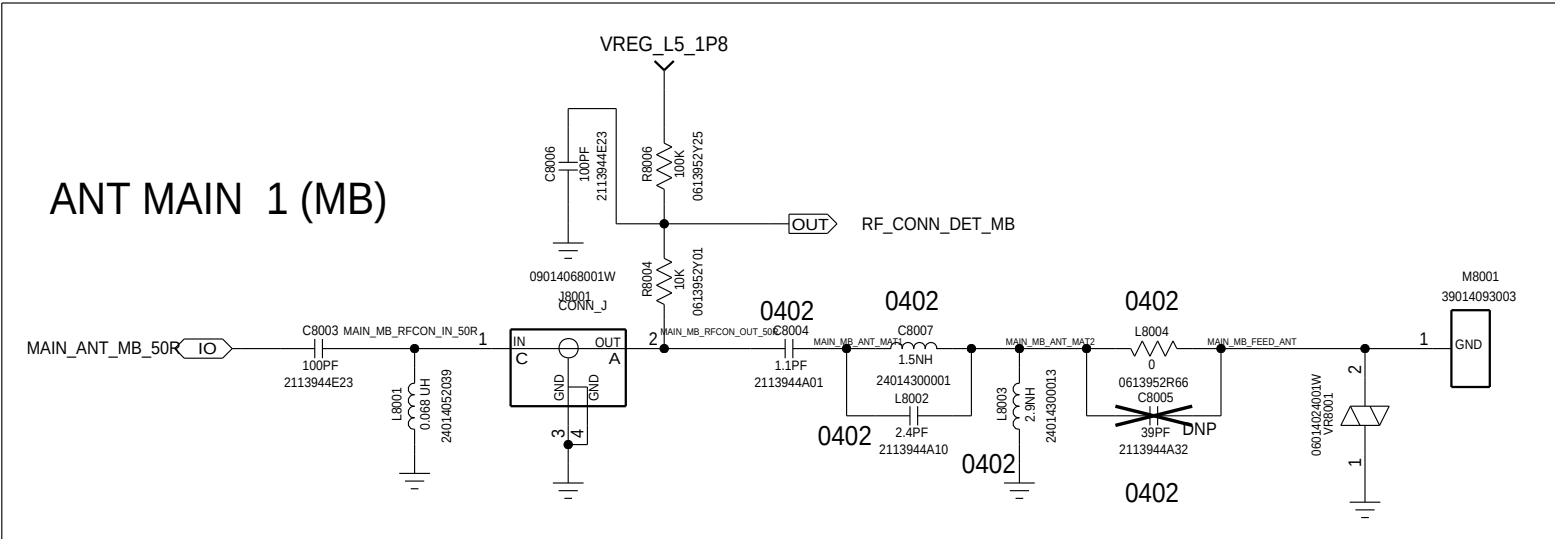
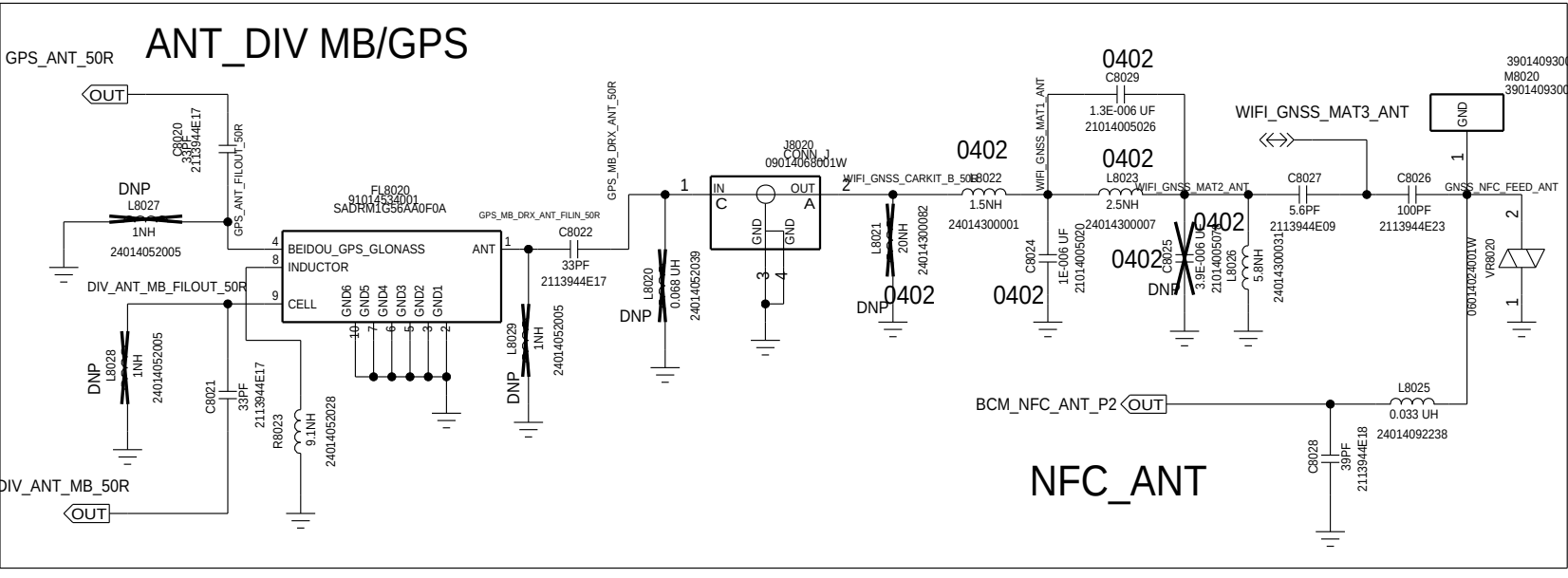
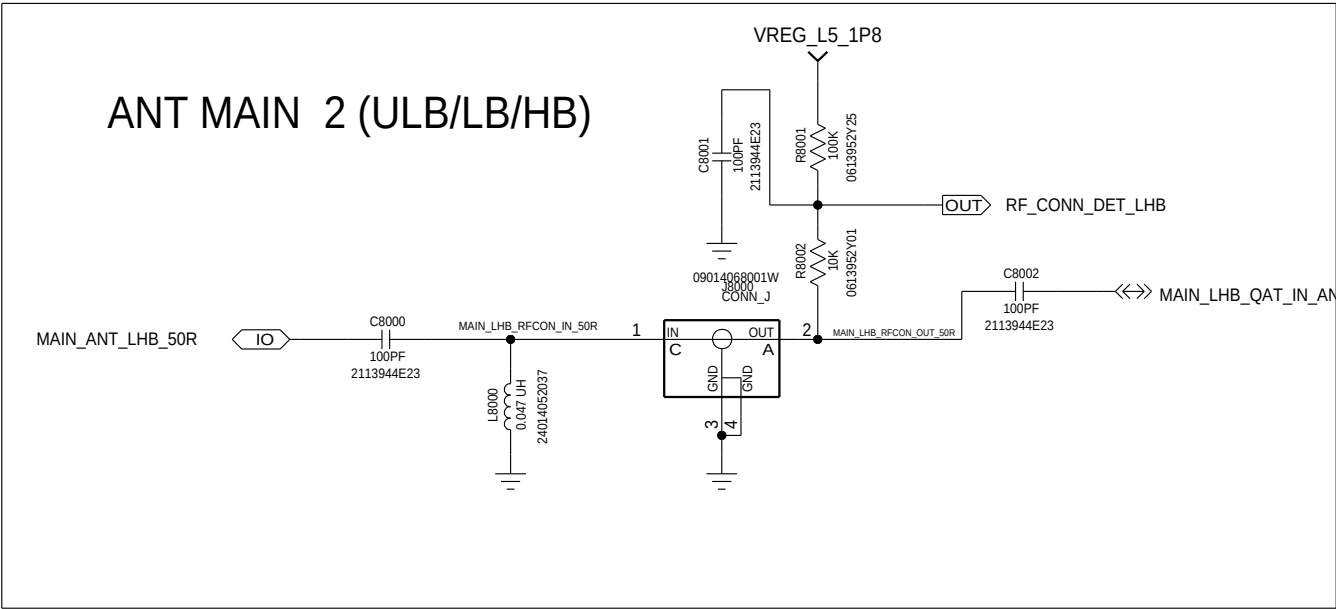


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Jan 05 14:29:33 2017		houdini_a	
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		B houdini_a.cpm	
		CAD FILE NAME:	
		PAGE 2 OF 64	
		REV 01	

8000:antennas

REF 8000-8049

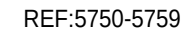
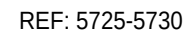
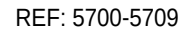


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		antennas	
Tue Feb 07 14:11:57 2017			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		PAGE 3 OF 64	

D

C

REF: 5700-5799



MOTOROLA
MOBILE DEVICES

ENGINEER: DENNIS BUDNICK

LAST MODIFIED BY: NATE CONNELL
Fri Dec 02 11:33:12 2016

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NOT IMPLY PUBLICATION

BLOCK TITLE:

audio_fe

	PROJECT NAME
--	--------------

B	houdini_a.cpm
---	---------------

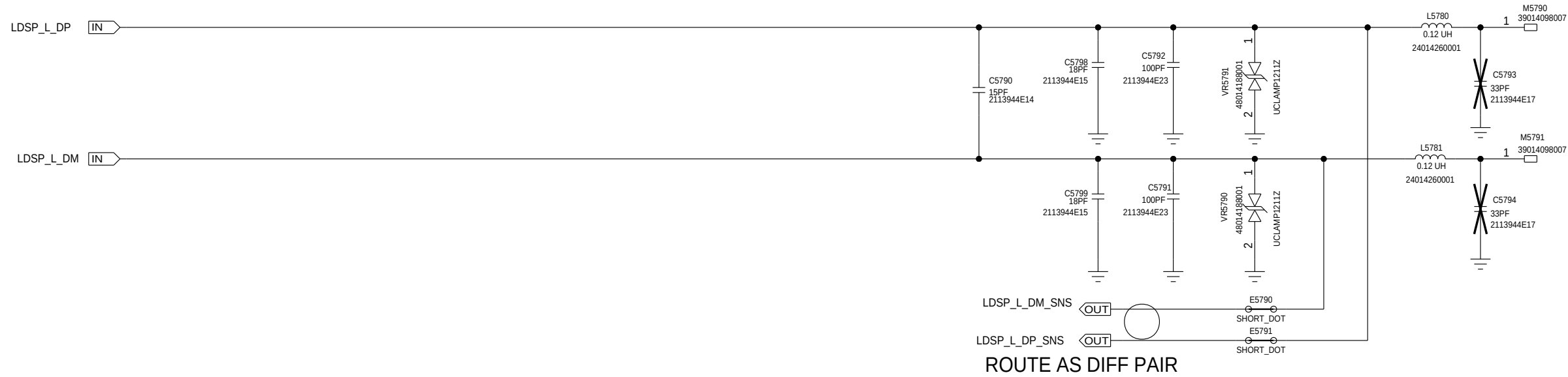
EV
1

CAD FILE NAME:

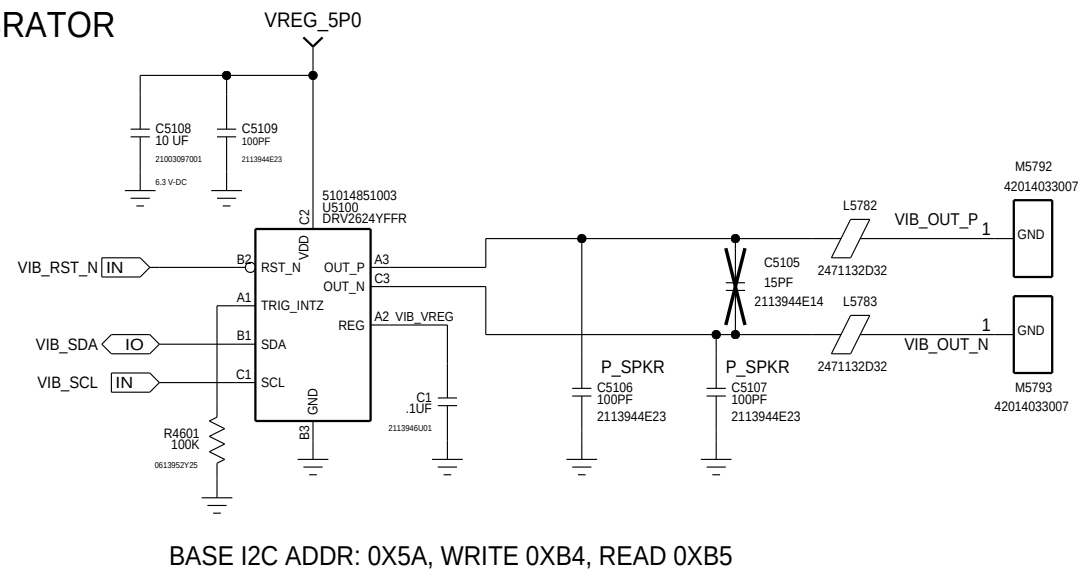
PAGE 5 OF 64

AFE: SPEAKERS

LOUDSPEAKER/EAR PIECE
REF:5790-5799



LINEAR VIBRATOR
REF 5100-5119

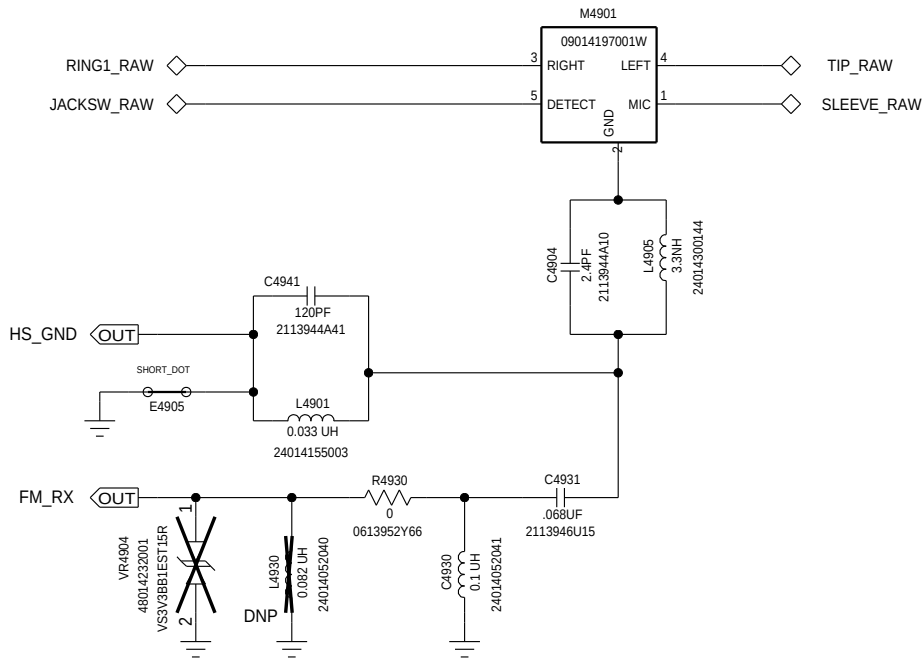
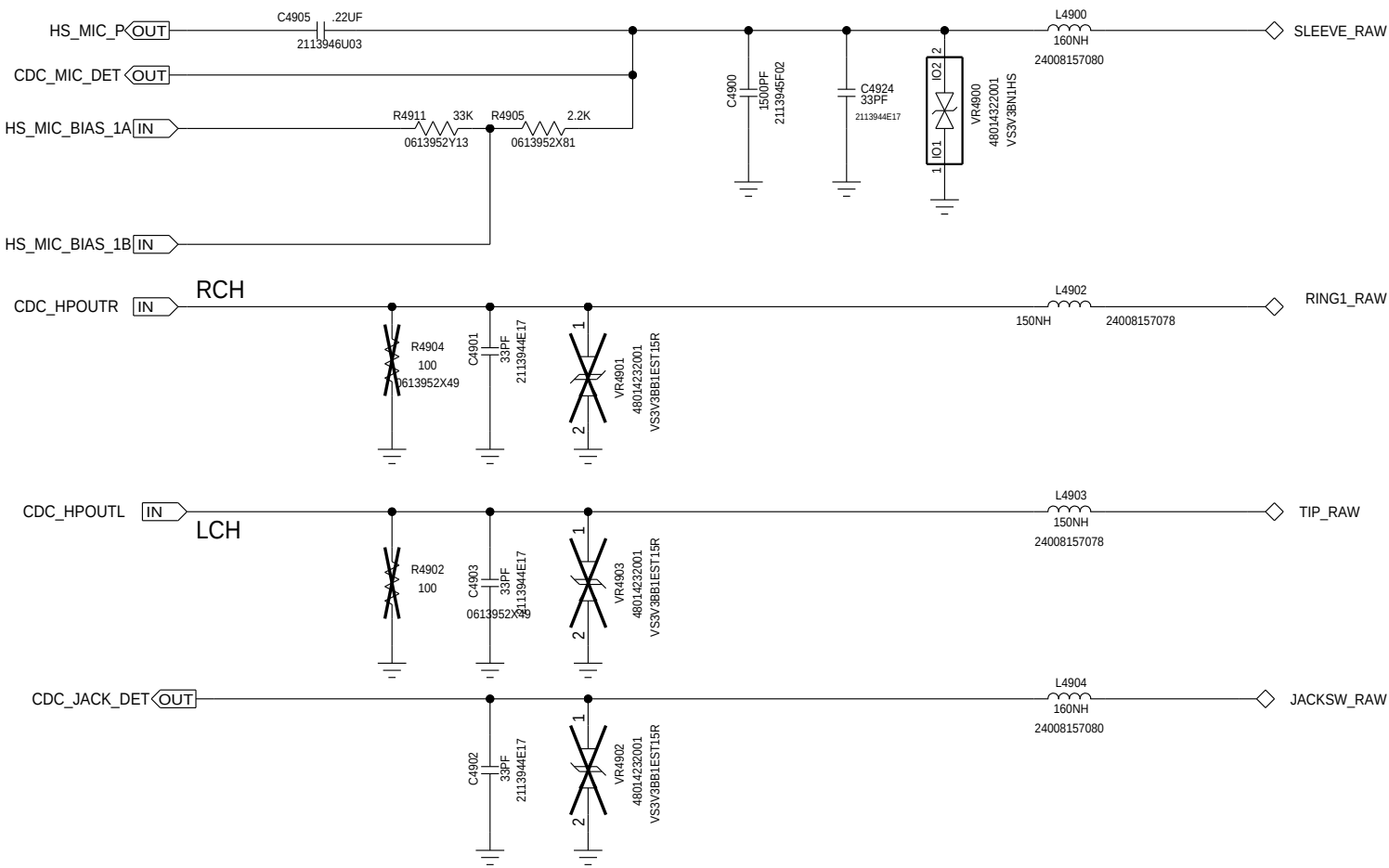


BASE I2C ADDR: 0X5A, WRITE 0XB4, READ 0XB5

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: DENNIS BUDNICK		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: HUNG PHAM Thu Nov 17 13:08:10 2016		PROJECT NAME houdini_a.cpm	
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		PAGE 6 OF 64	

AFE: 3.5MM HEADSET

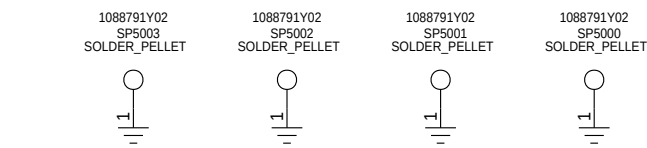
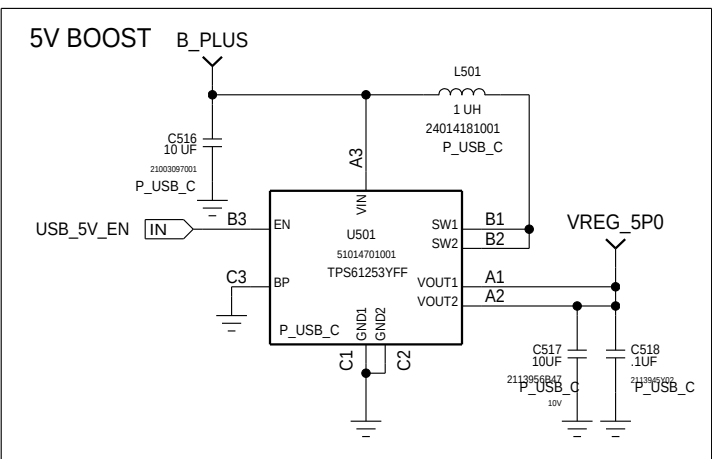
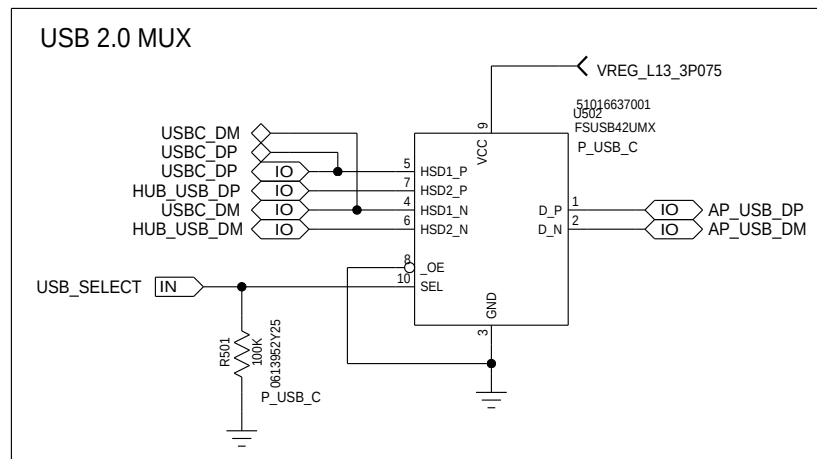
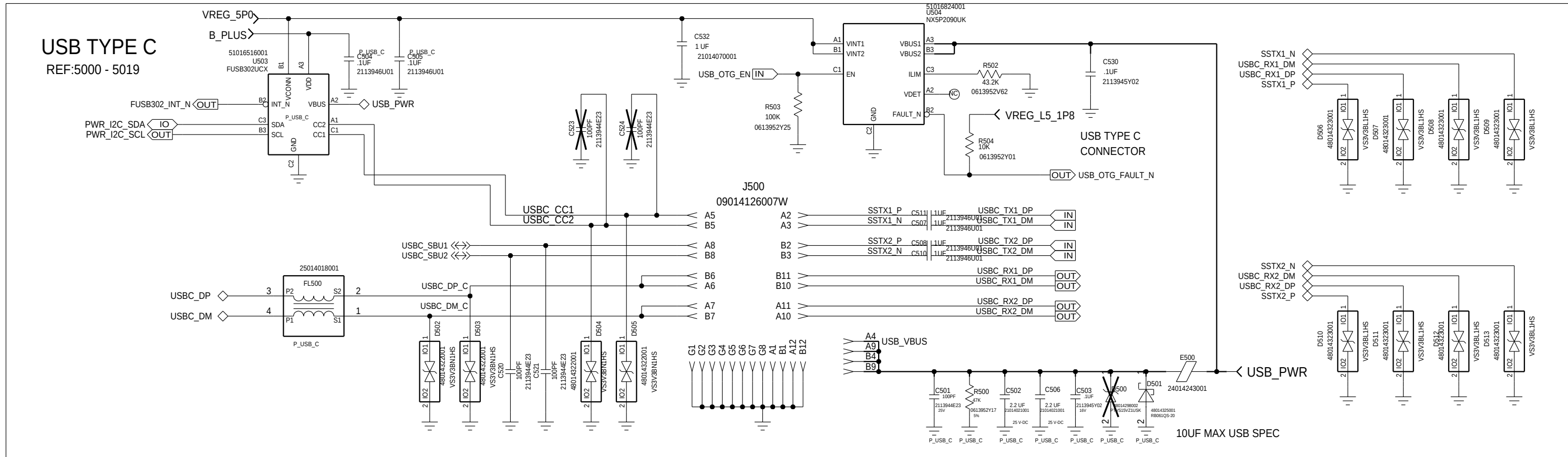
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MMI Tue Nov 15 13:46:18 2016		PROJECT NAME houdini_a.cpm	
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		CAD FILE NAME: PAGE 7 OF 64	

PWR: USB-C

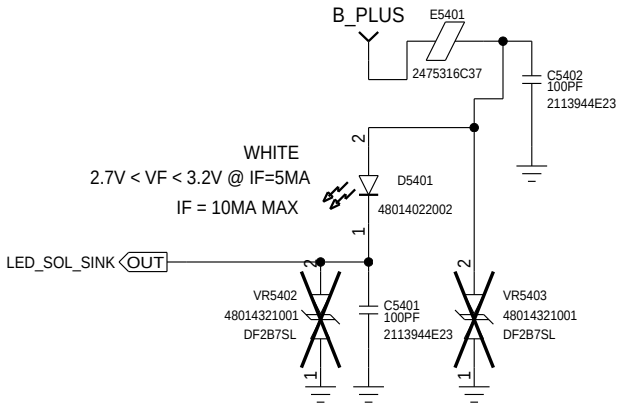
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APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: NATE CONNELL Tue Feb 07 11:38:41 2017		PROJECT NAME houdini_a.cpm	
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REV 01			

PWR: LIGHTING

SOL LED



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

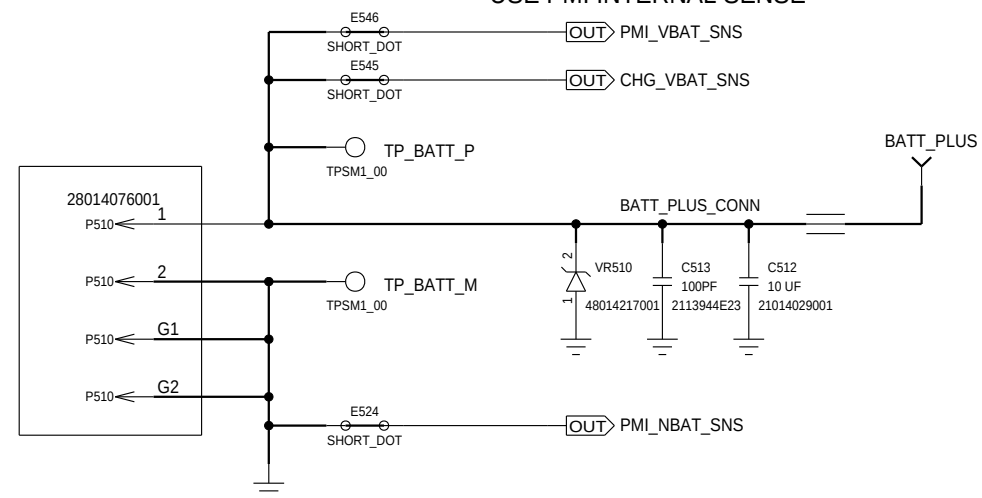
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: TIM MILLER Tue Nov 15 13:46:15 2016		power	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B houdini_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 9 OF 64	

PWR: BATT

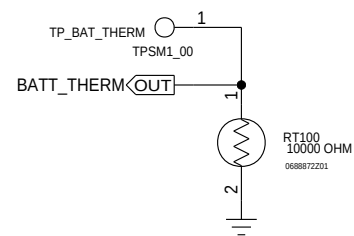
REF:510 - 550

BATTERY BLOCK

USE PMI INTERNAL SENSE

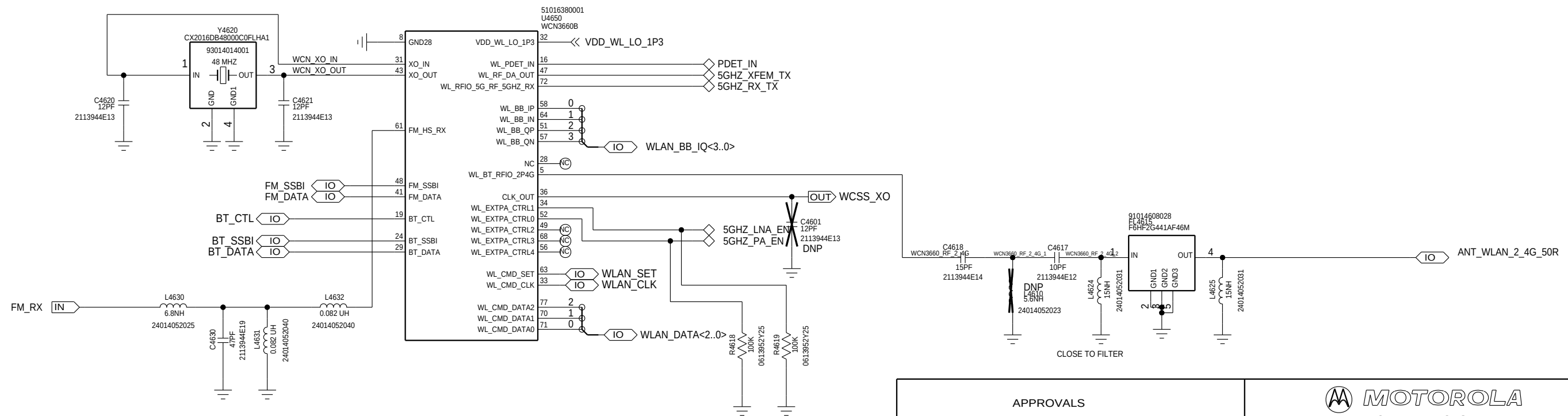
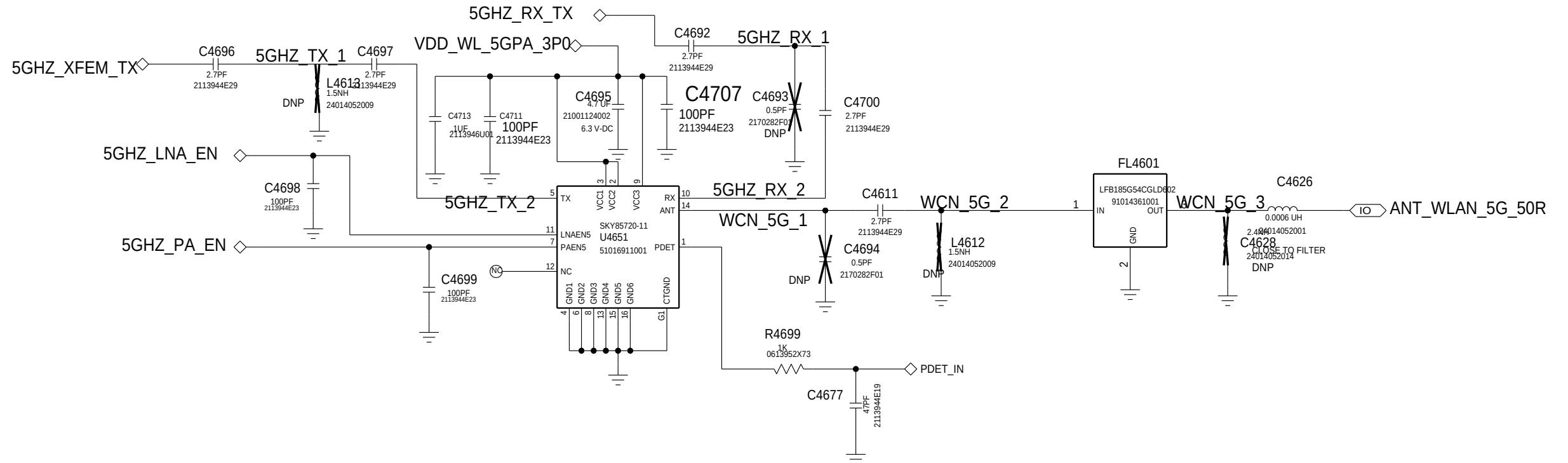


BATT THERMISTOR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG	BLOCK TITLE: power		
LAST MODIFIED BY: NATE CONNELL Tue Dec 13 16:58:42 2016			
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		CAD FILE NAME:	PAGE 10 OF 64

WCN3660B RF/CTRL
REF 4600-4699

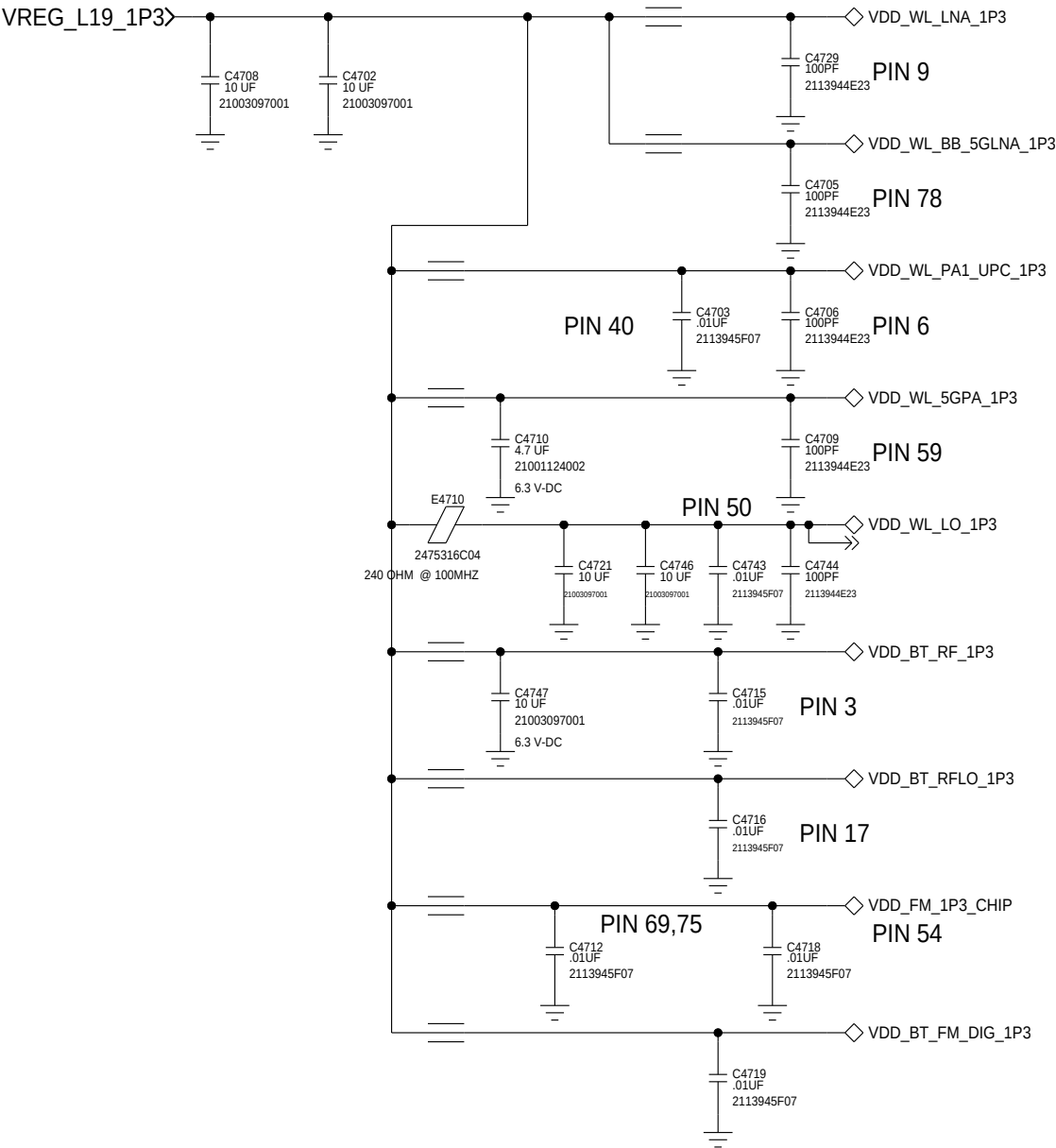


BASED ON 80-WL006-44 REV C

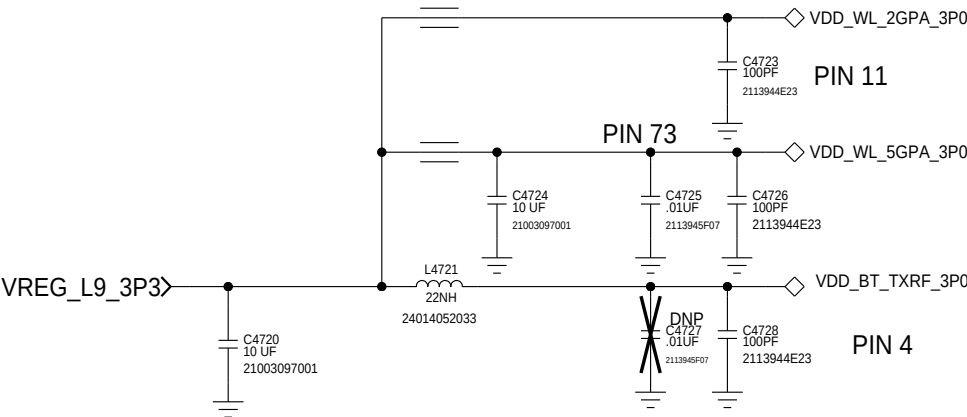
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: connectivity		
LAST MODIFIED BY: FROEHLING Fri Dec 16 14:34:31 2016			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B houdini_a.cpm	REV 01
		CAD FILE NAME:	PAGE 11 OF 64

WCN3660B POWER
REF 4700-4799

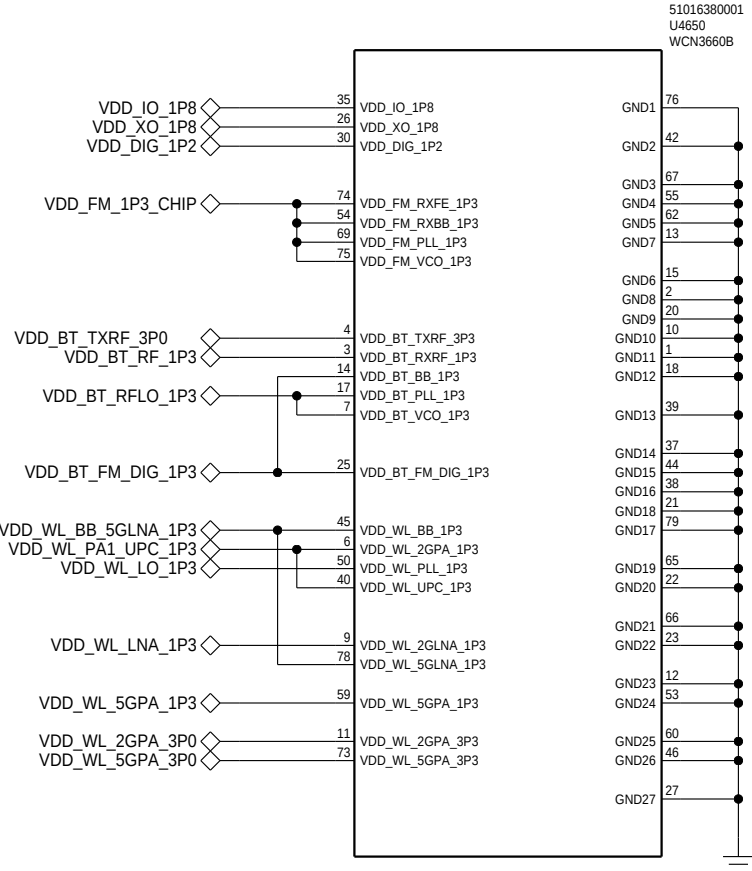
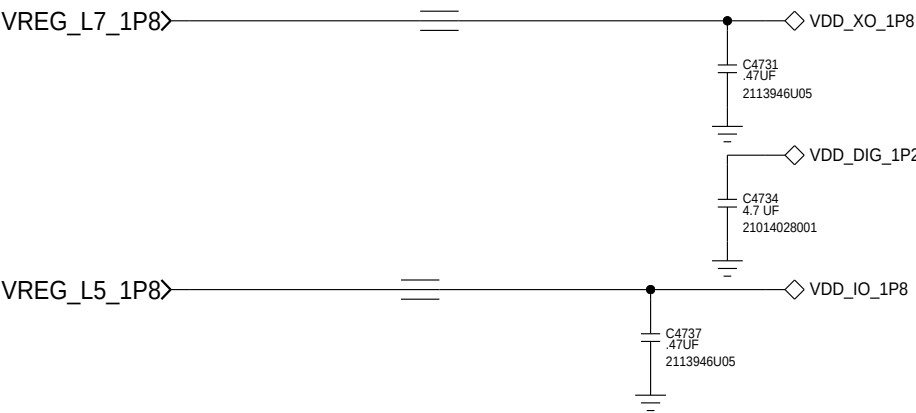
BT_FM_WLAN_1P3



WLBT_3.0V



1.8V/2.2V



BASED ON 80-WL006-44 REV C

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue Nov 15 13:46:10 2016		connectivity	
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		CAD FILE NAME: PAGE 12 OF 64	

A



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING	BLOCK TITLE: connectivity		
LAST MODIFIED BY: FROEHLING Tue Dec 13 17:00:08 2016			
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		CAD FILE NAME:	PAGE 13 OF 64

USR: FPC

REF: 6600 - 6699

8

7

6

5

4

3

2

1

D

C

B

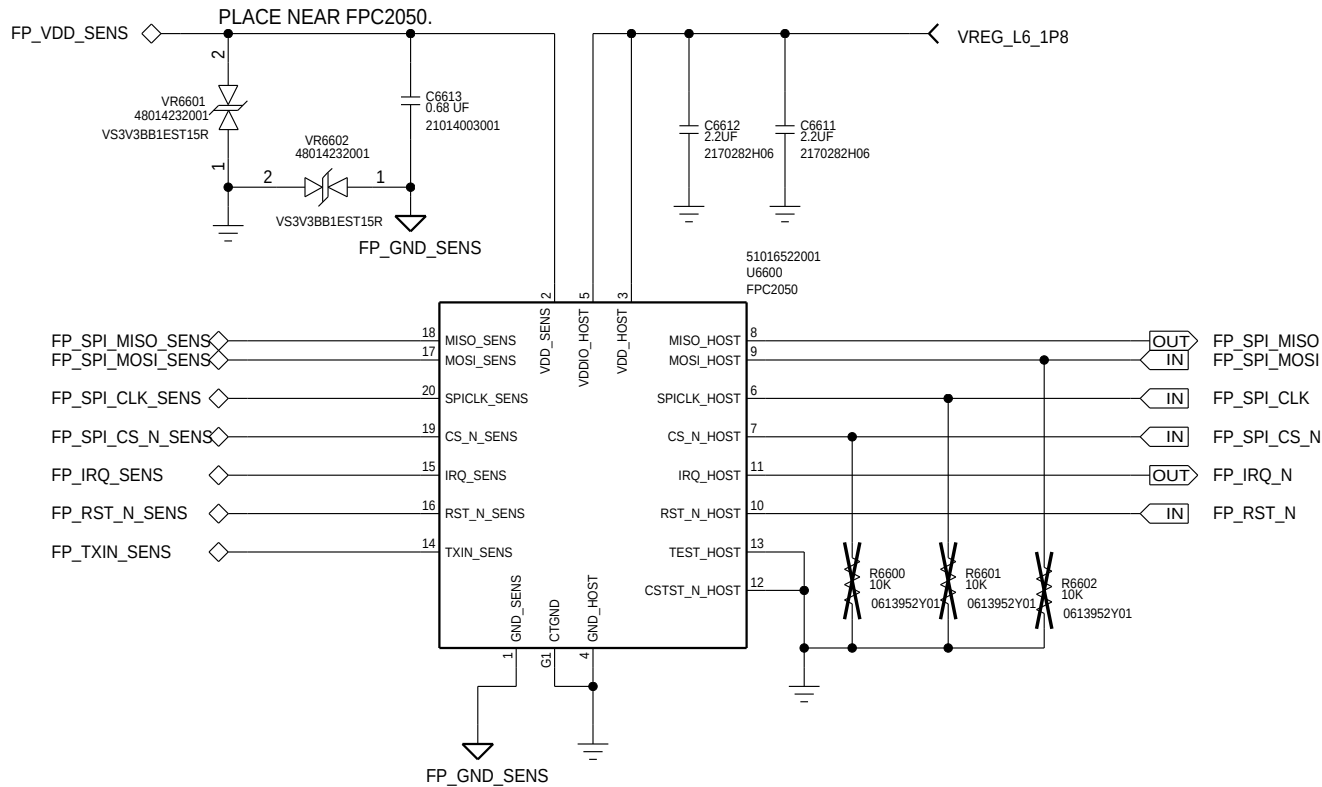
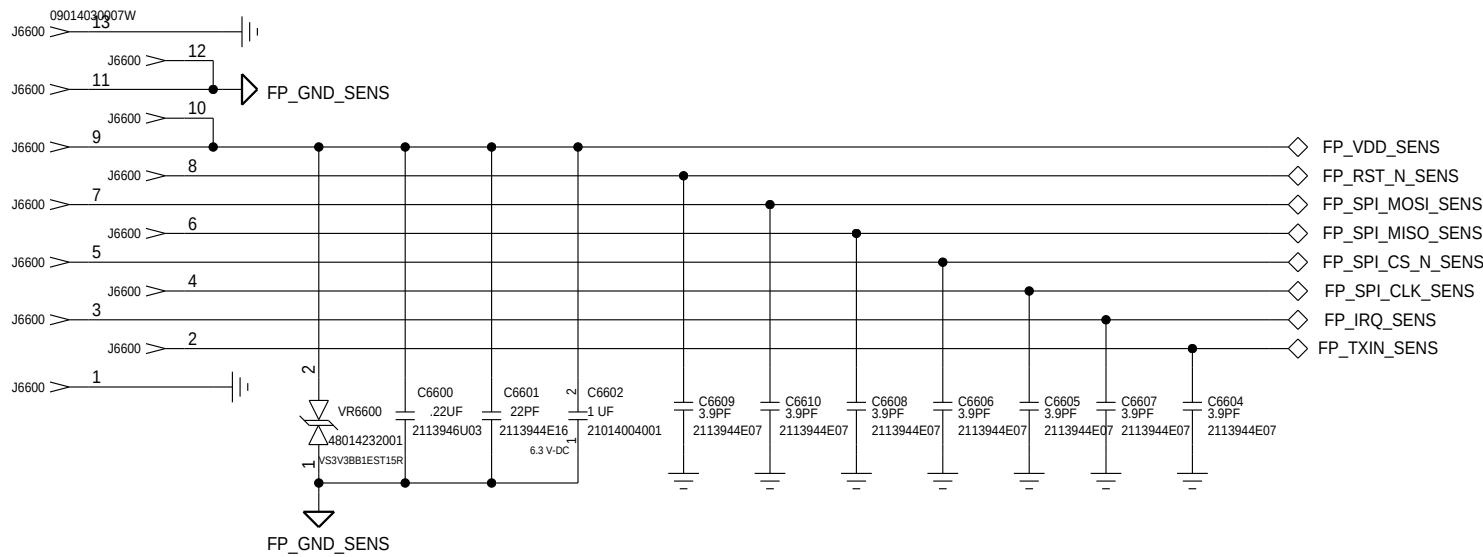
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D

C

B

A



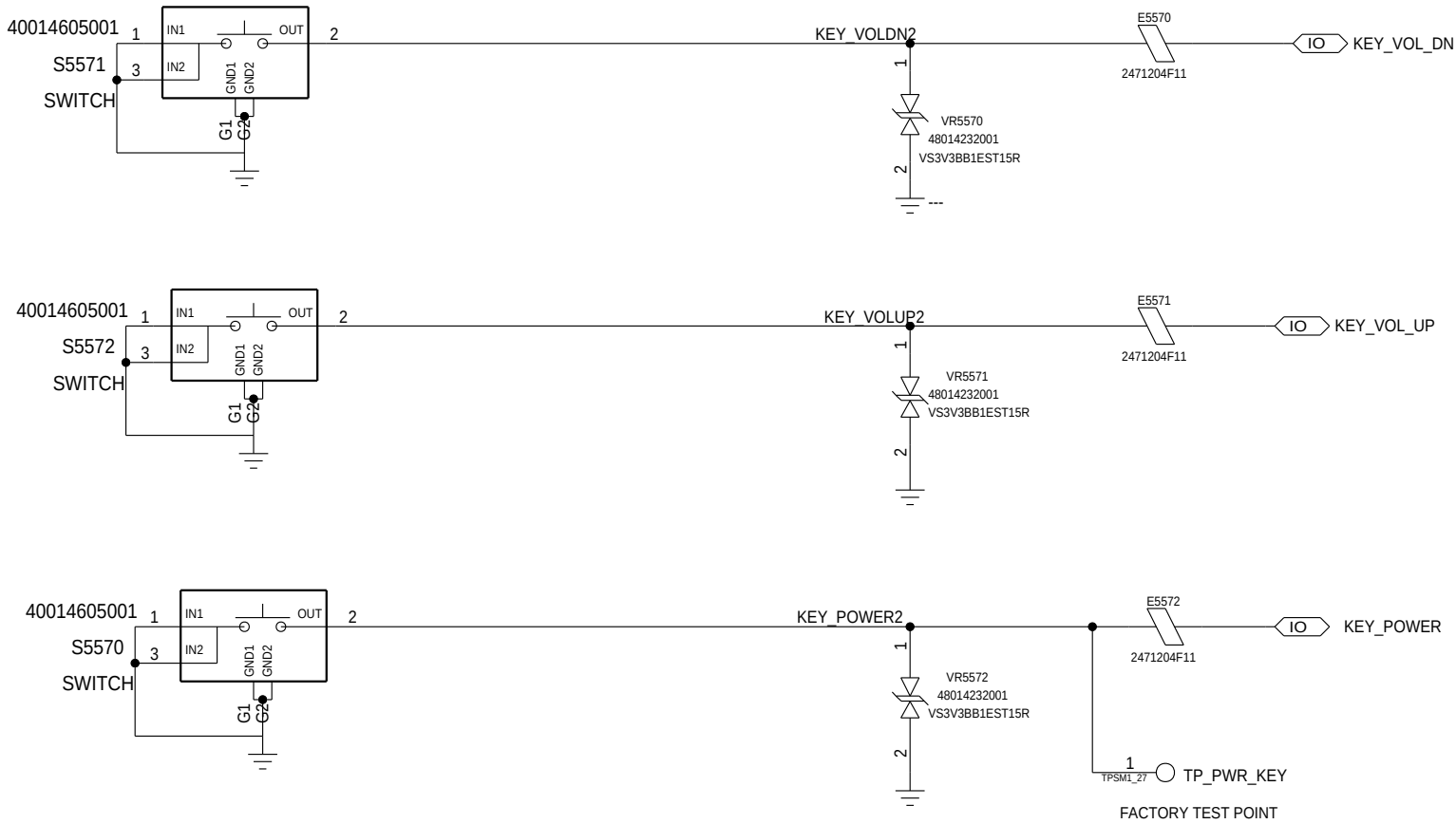
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ENGINEER: NATE CONNELL		BLOCK TITLE:	
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		B houdini_a.cpm	
		REV 01	
		PAGE 14 OF 64	

SIDE KEY

REF:5570-5579

MECHANICAL KEYS

VOL_UP/DOWN/GND TPS NEED P0 BRINGUP ONLY



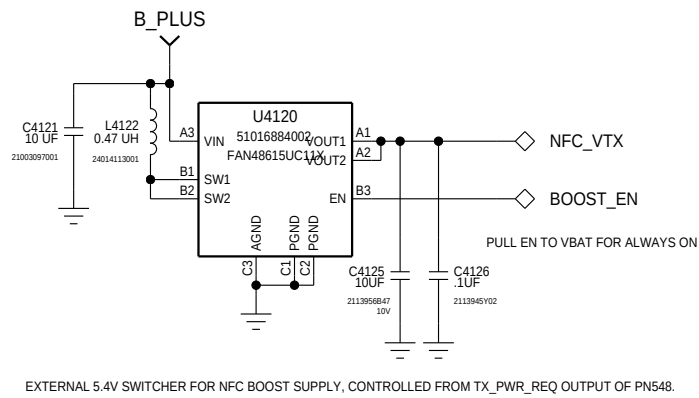
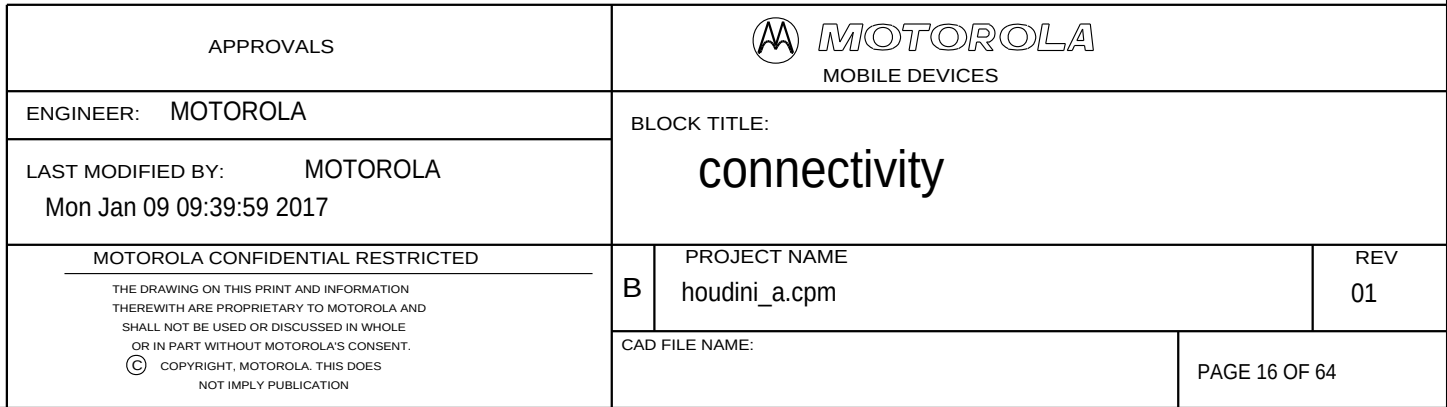
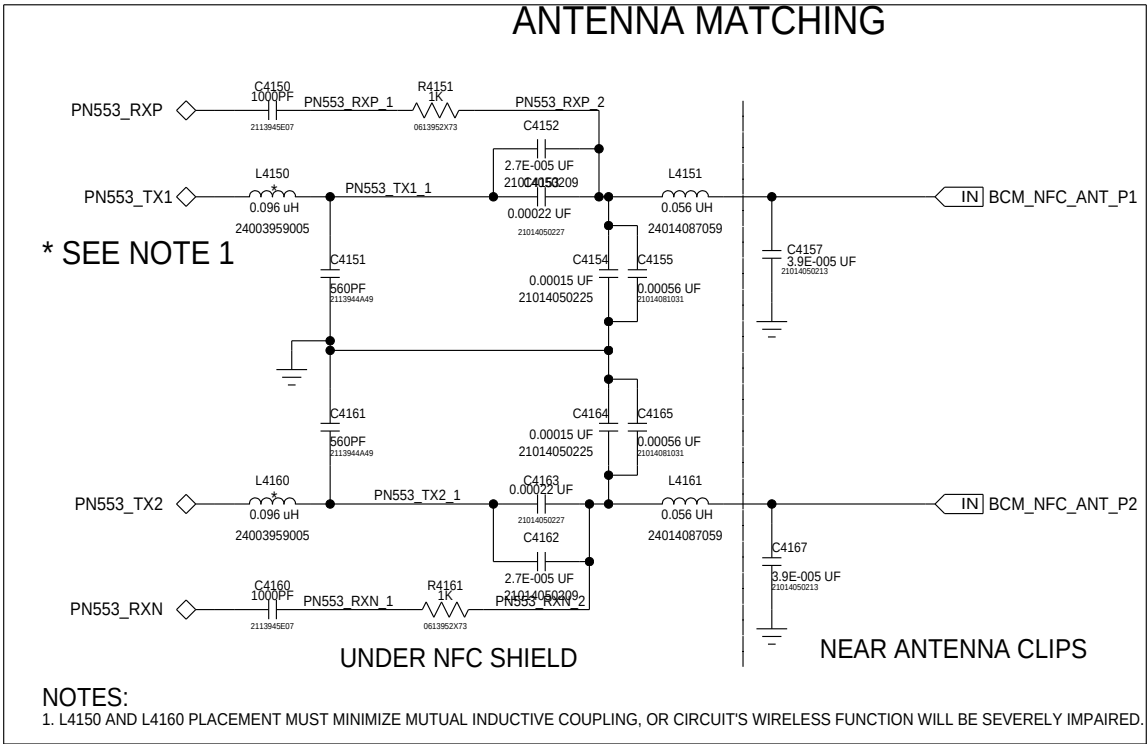
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue Nov 15 13:46:12 2016		connectivity	
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		B houdini_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 15 OF 64	

D

C

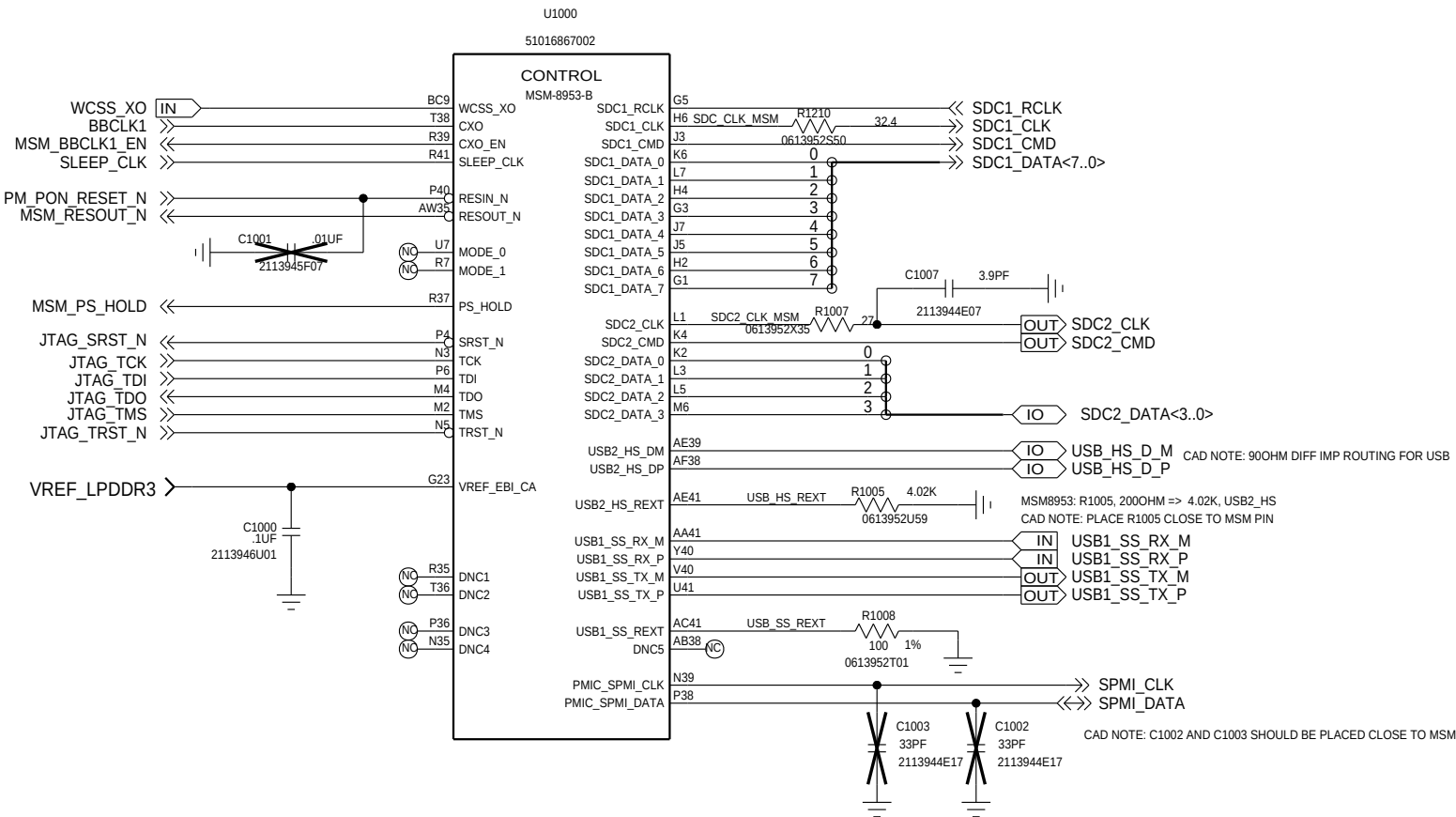
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A



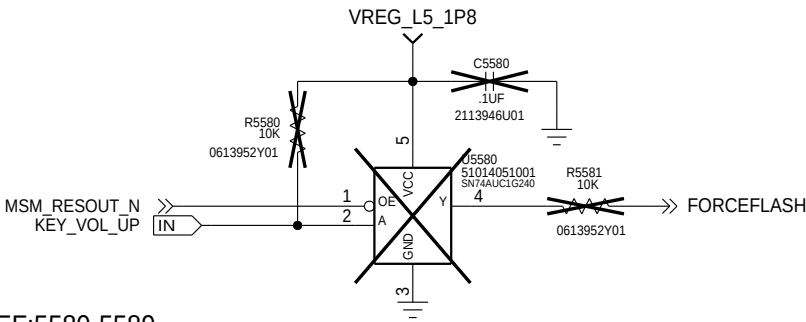
MSM: CTRL

REF 1000-1099



FORCE FLASH SUPPORT WITH VOL UP KEY

(DNI FOR PRODUCTION)



REF:5580-5589

APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Tue Nov 15 13:45:58 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

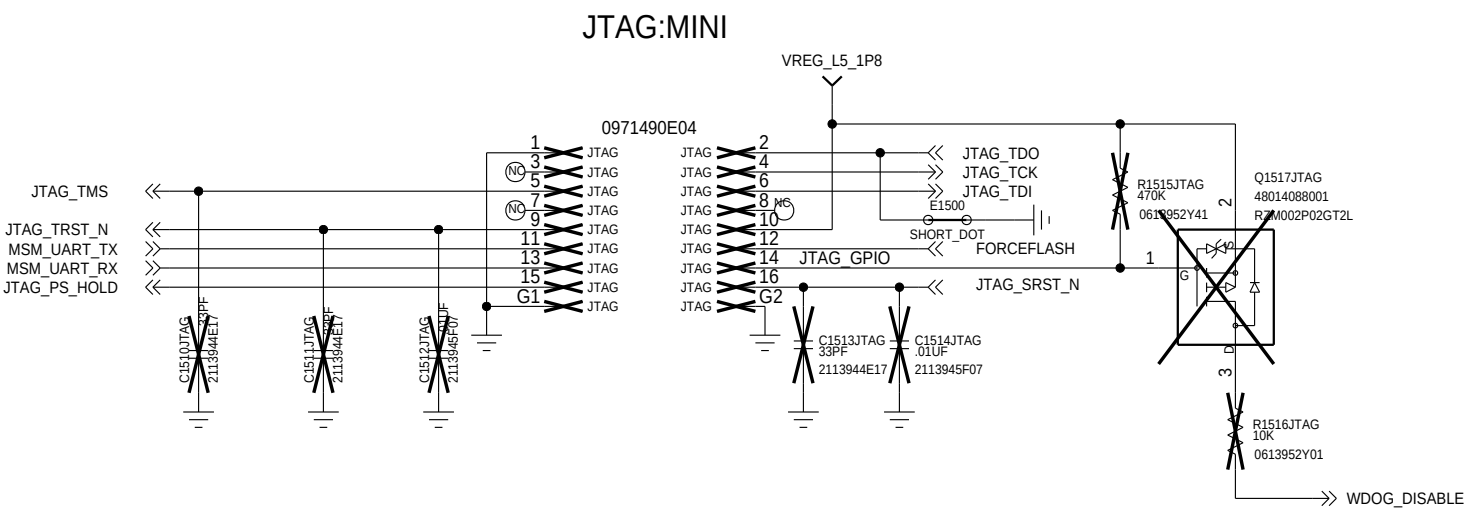
PROJECT NAME
houdini_a.cpm

CAD FILE NAME:

REV
01

MSM: DEBUG

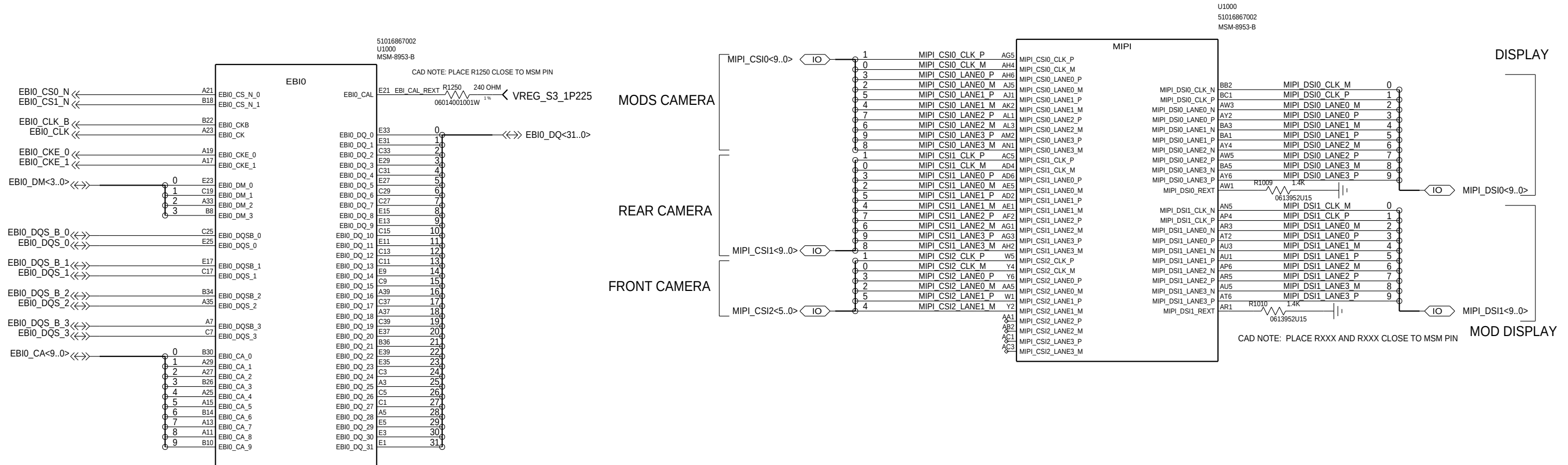
REF 1000-1099



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Feb 07 14:59:47 2017		ap_pmic	
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		B	houdini_a.cpm
		CAD FILE NAME:	
		PAGE 18 OF 64	
REV 01			

MSM: MIPI/DSI/CSI/EBI0

REF 1200-1299



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: Tue Nov 15 13:45:59 2016			
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		CAD FILE NAME:	PAGE 19 OF 64

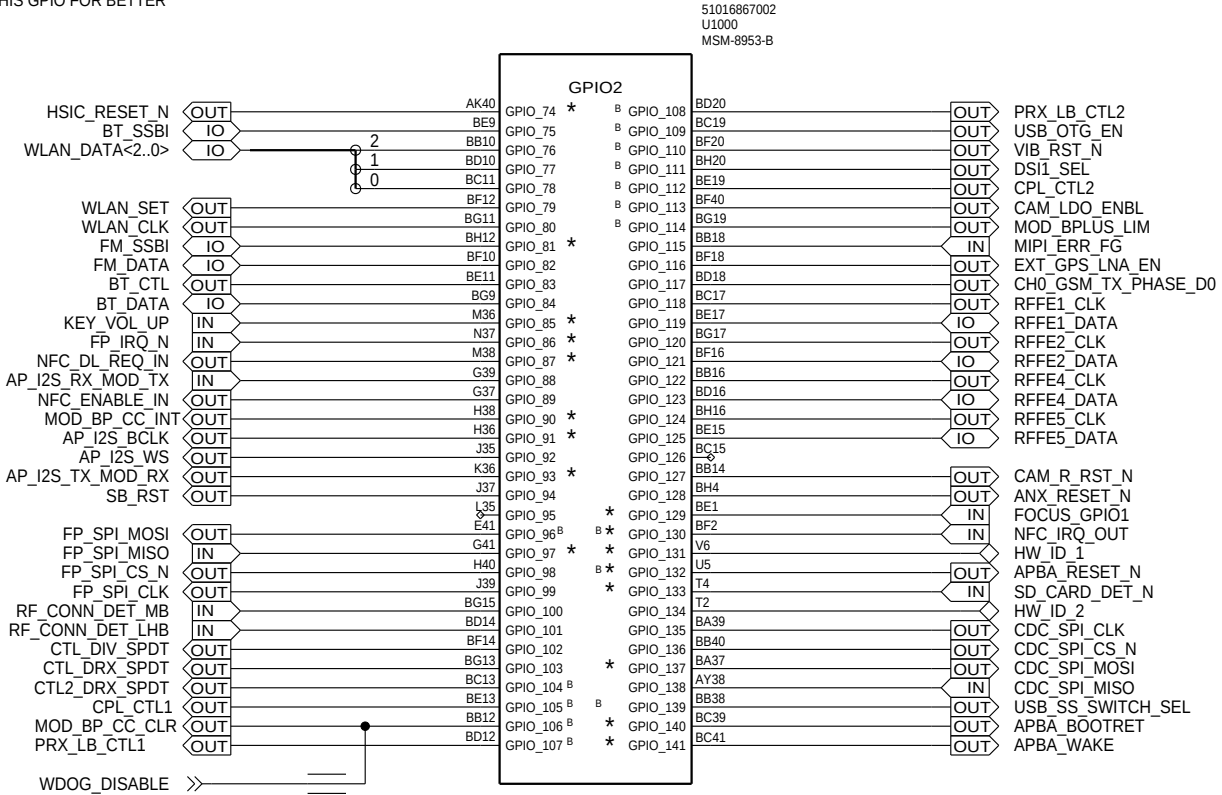
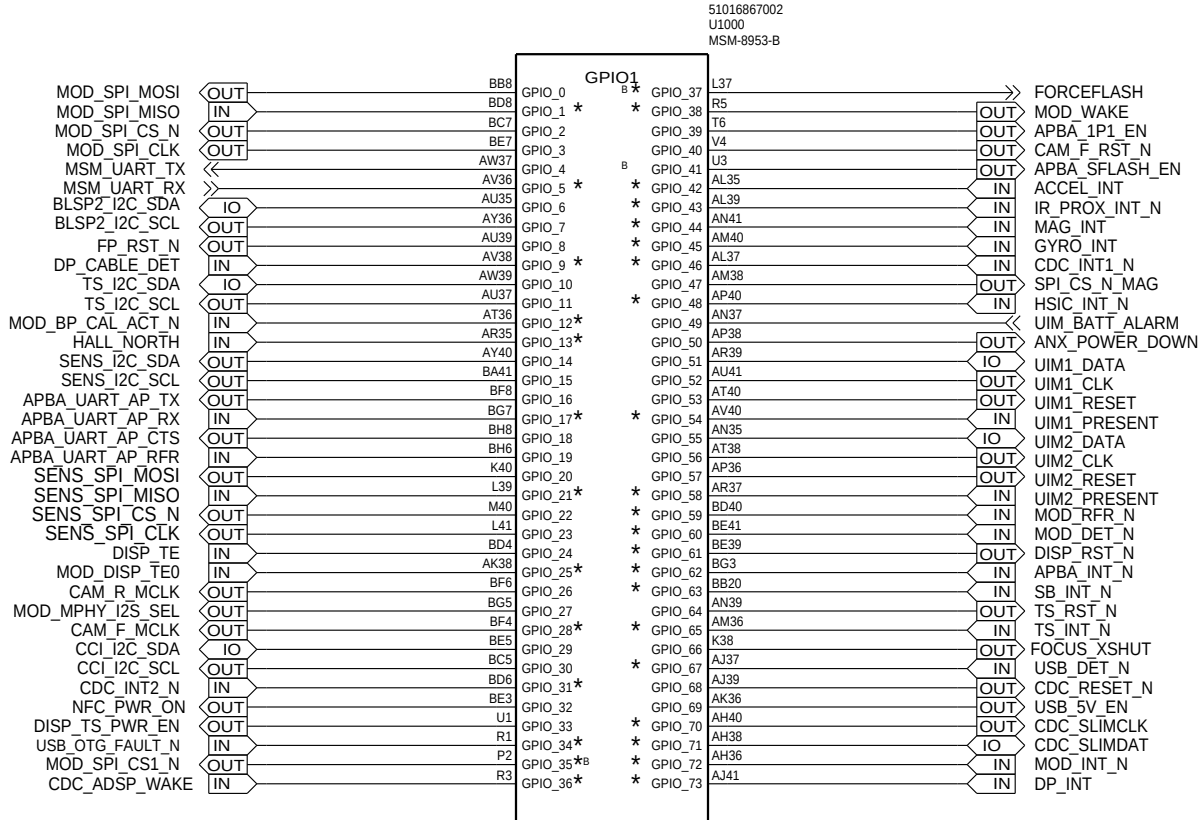
MSM: GPIO (0-141)

REF 1200-1299

NOTE: ASTERISK(*) INDICATE MODEM POWER MANAGEMENT (MPM) WAKE-UP PINS.

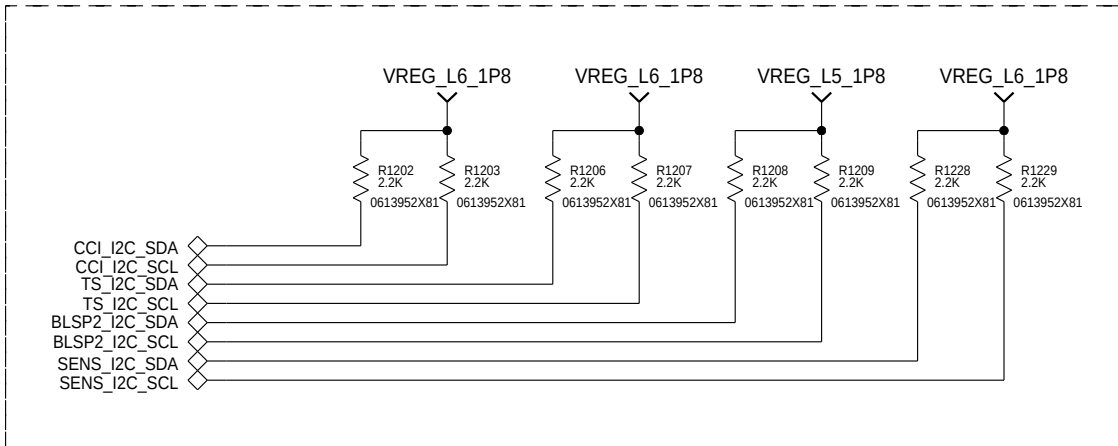
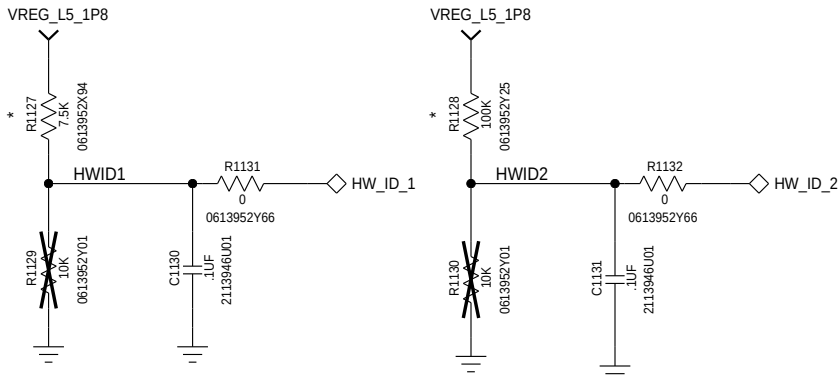
NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

NOTE: RESERVE GPIO_106 FOR WDOG_DISABLE FUNCTION.
DO NOT HAVE A PULL-UP ON THIS GPIO FOR BETTER SW DEBUG CAPABILITY.



MOT BOARD ID

FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>

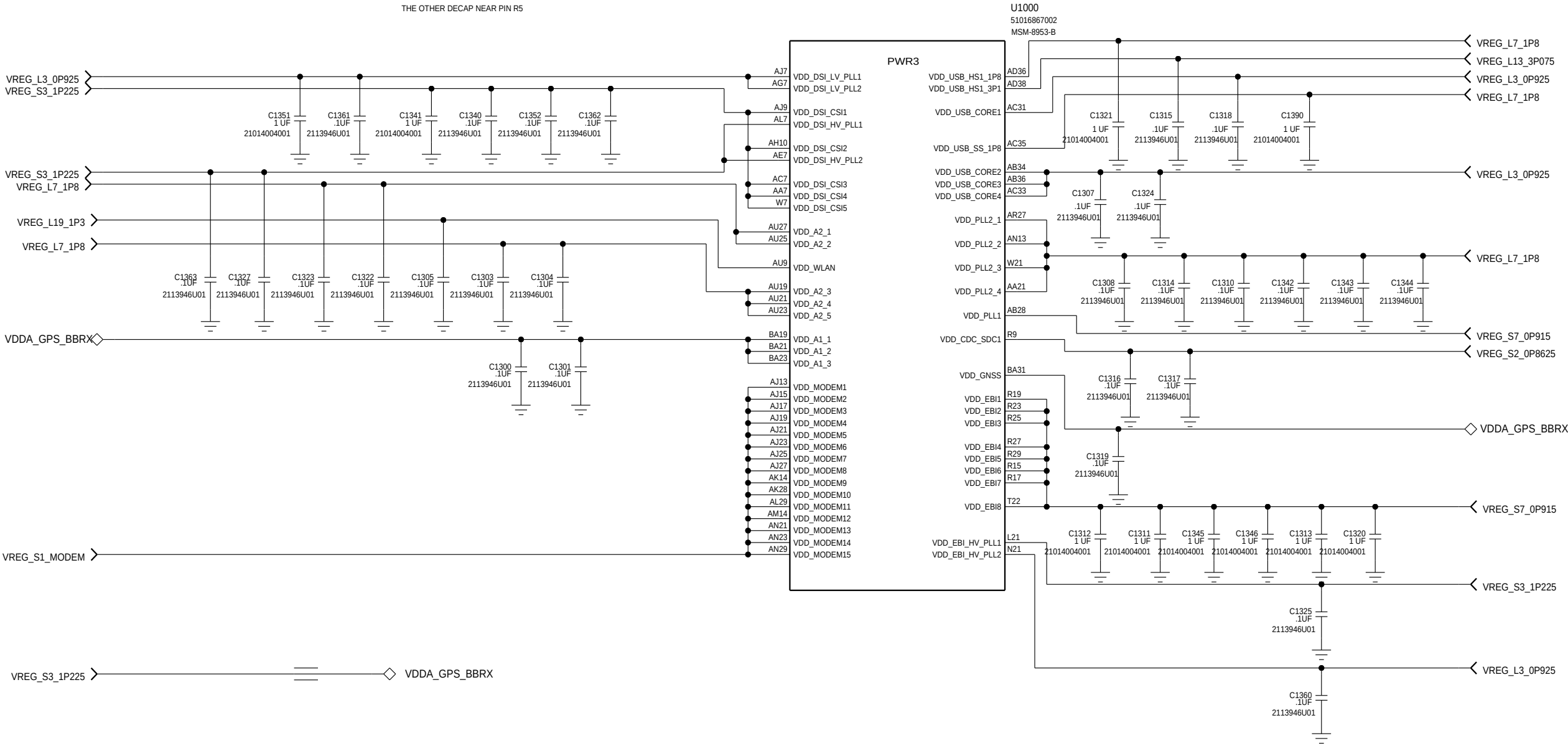


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Nov 15 13:46:00 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 20 OF 64	

MSM: PWR1

REF 1300-1399

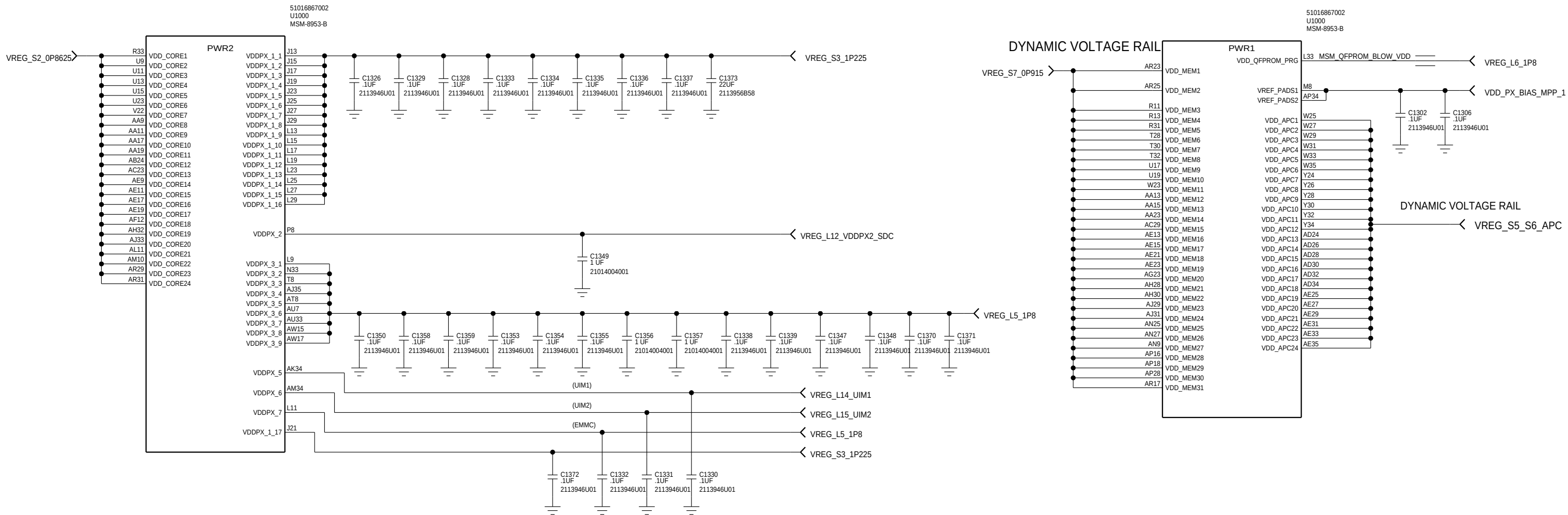
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Nov 15 13:46:01 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 21 OF 64	

MSM: PWR2

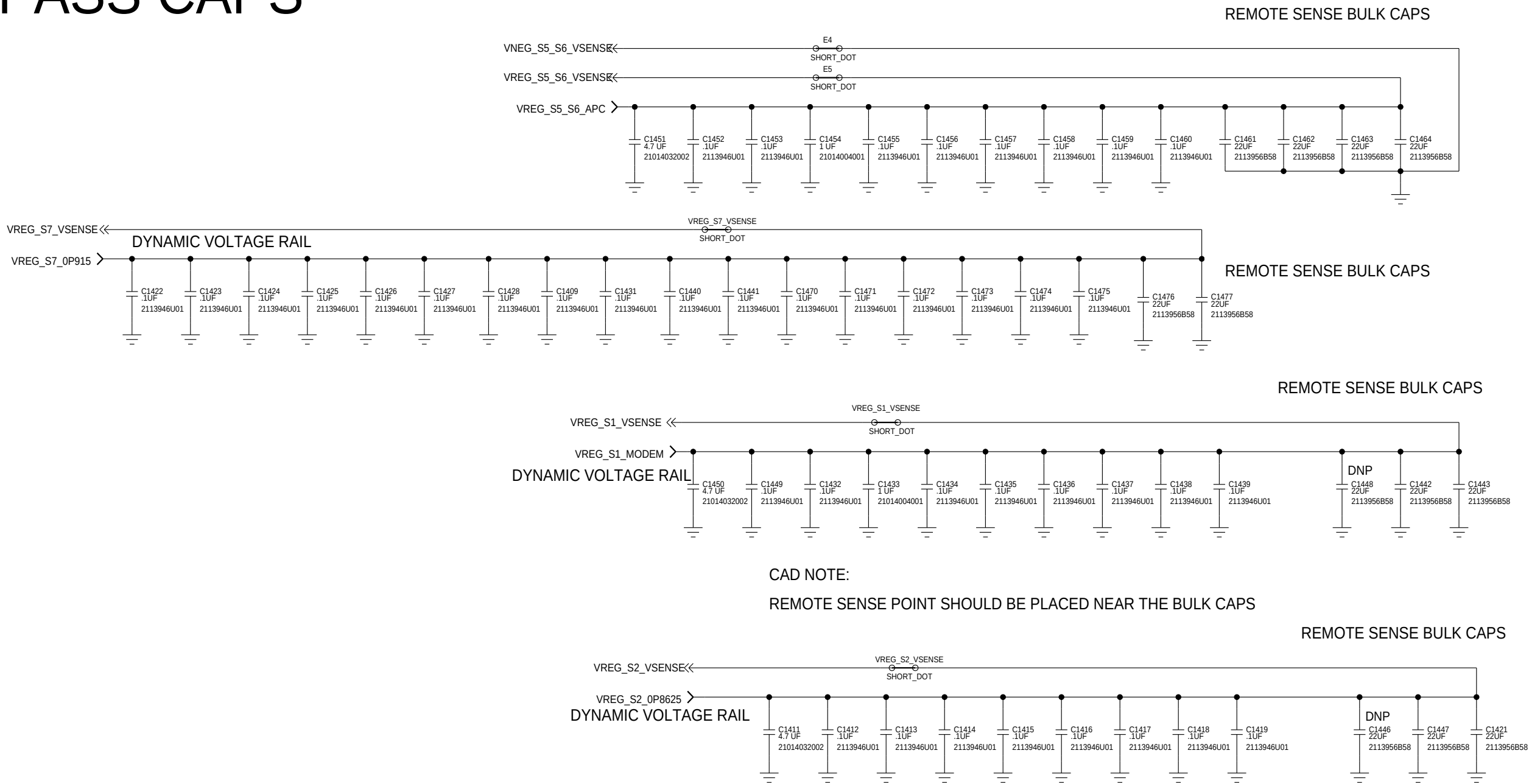
REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Nov 15 13:46:01 2016		ap_pmic	
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		B	houdini_a.cpm
		REV 01	
		PAGE 22 OF 64	

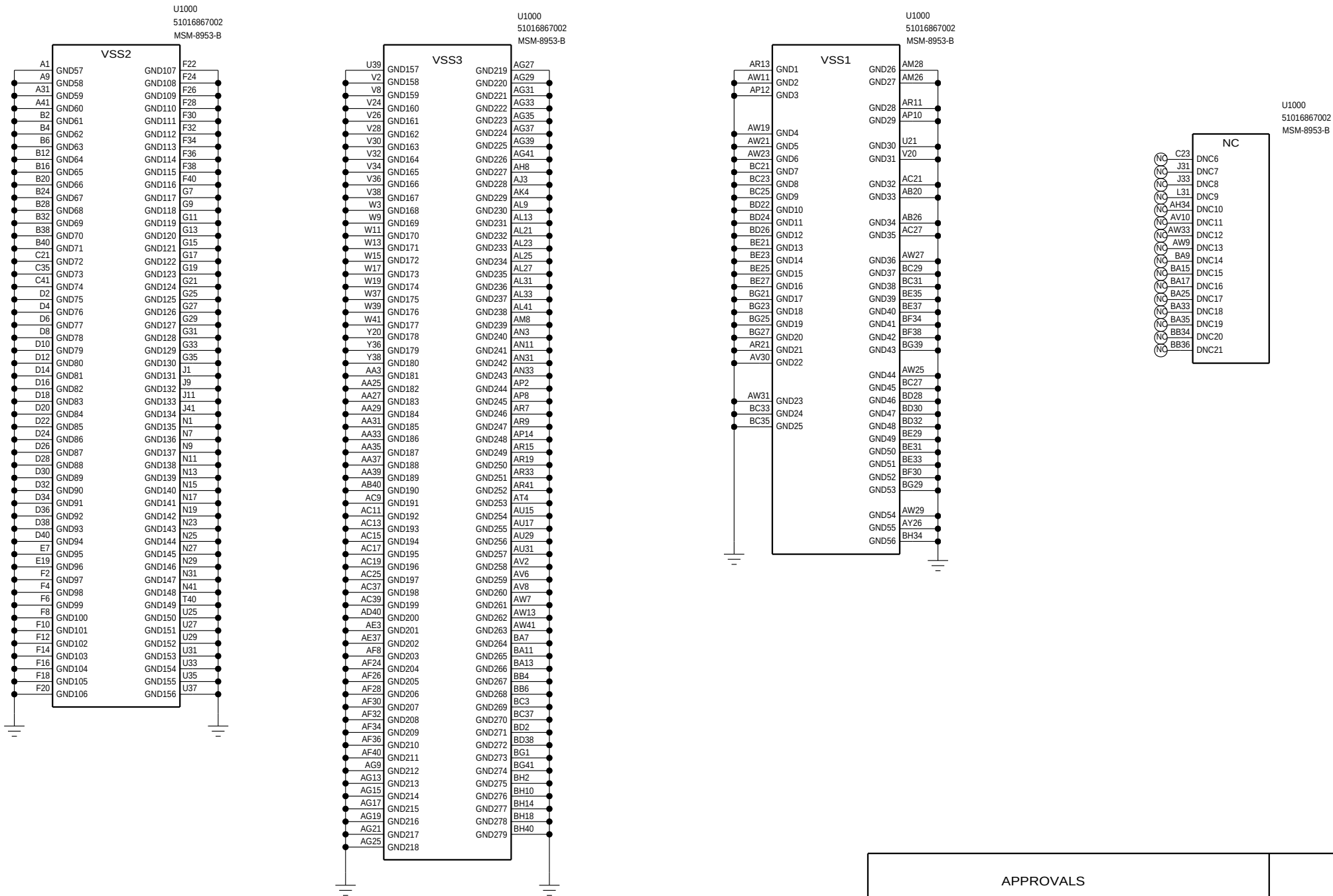
MSM: BYPASS CAPS

REF 1400-1499



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: FAN021 Tue Nov 15 13:46:02 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 23 OF 64	

MSM: GROUND/DNC

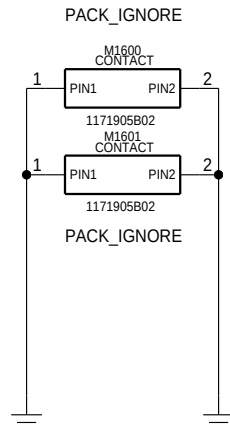
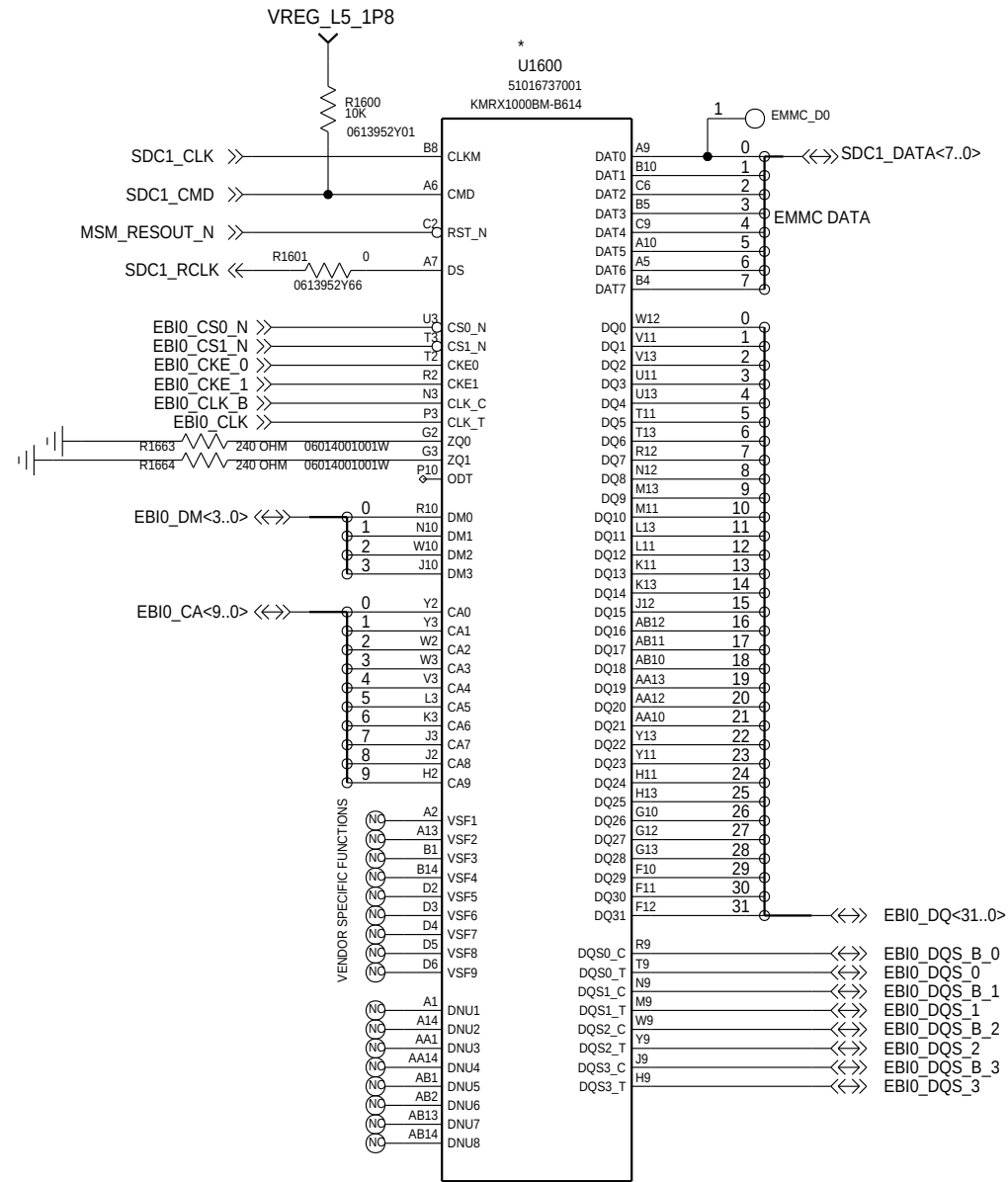
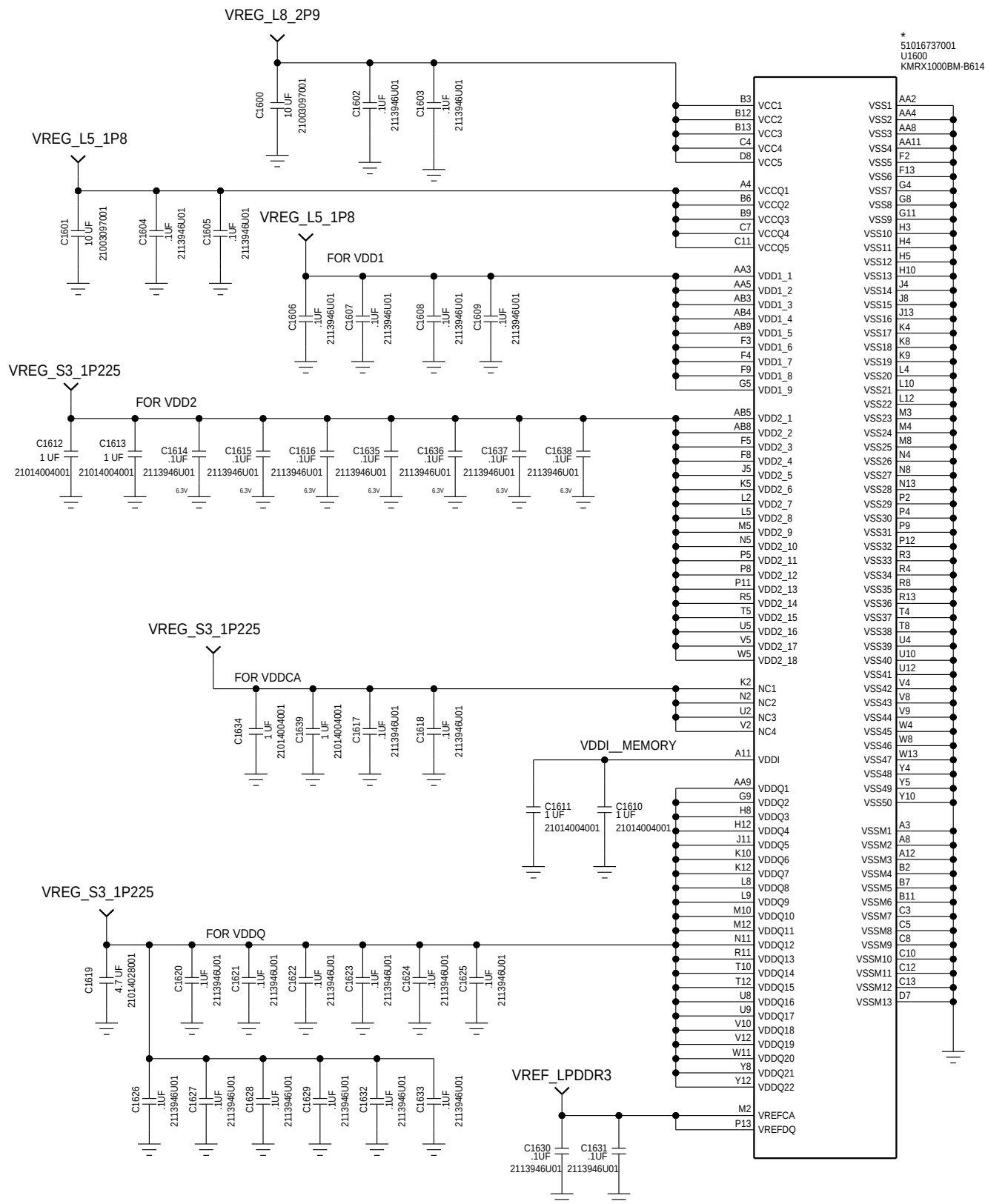


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
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		B houdini_a.cpm	
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MEMORY: EMCP

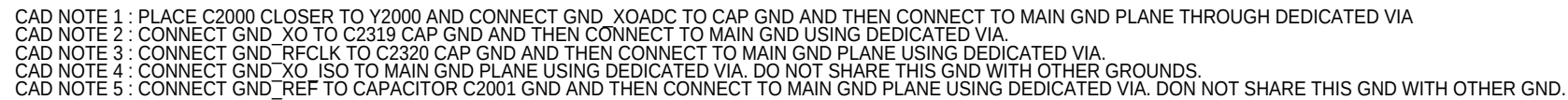
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MCP: 16GB EMMC, 2GB LPDDR3



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Tue Nov 15 13:46:04 2016		ap_pmic	
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		REV 01	
		PAGE 25 OF 64	

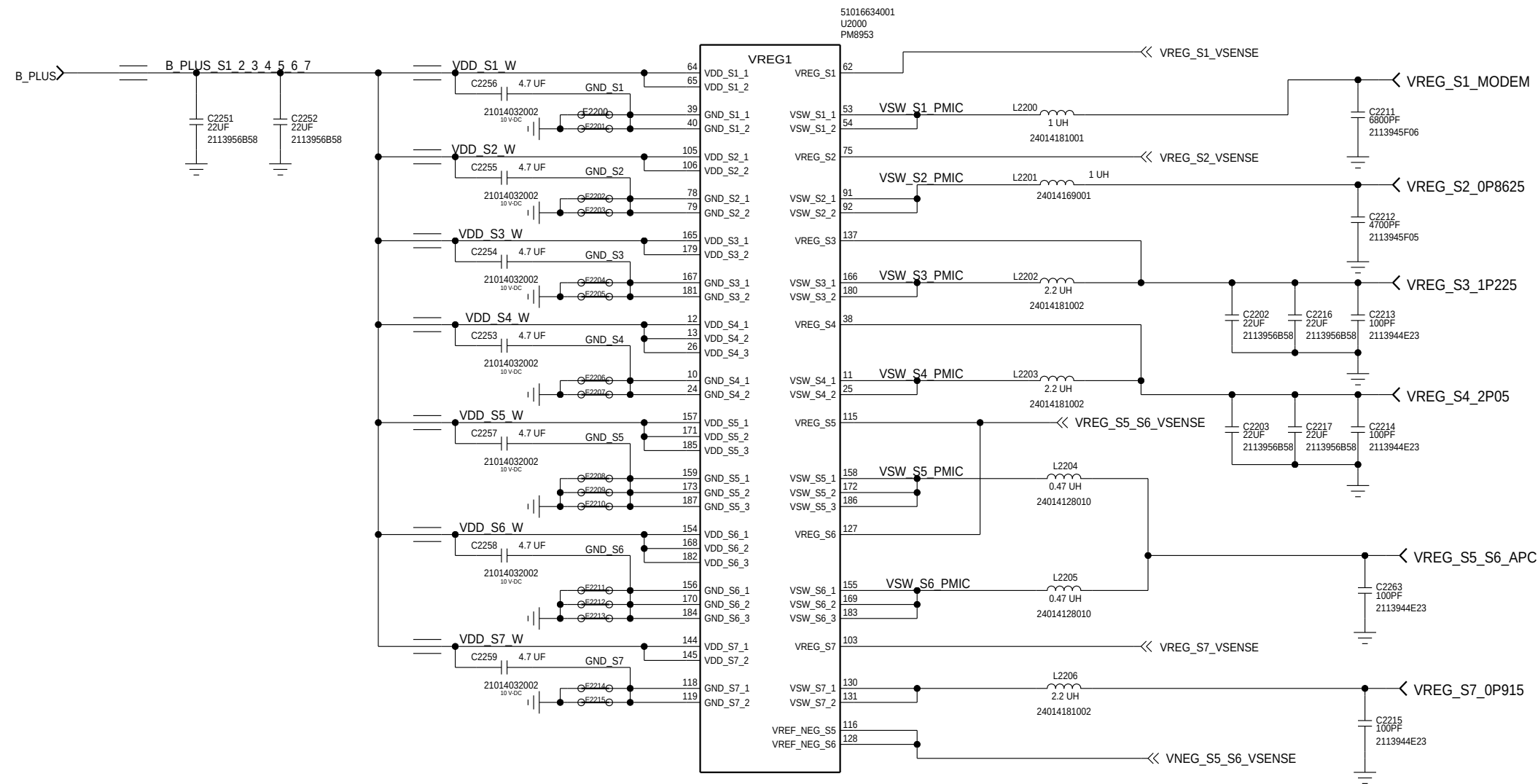
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	PAGE 26 OF 64

PM8953: SMPS

REF: 2200-2299

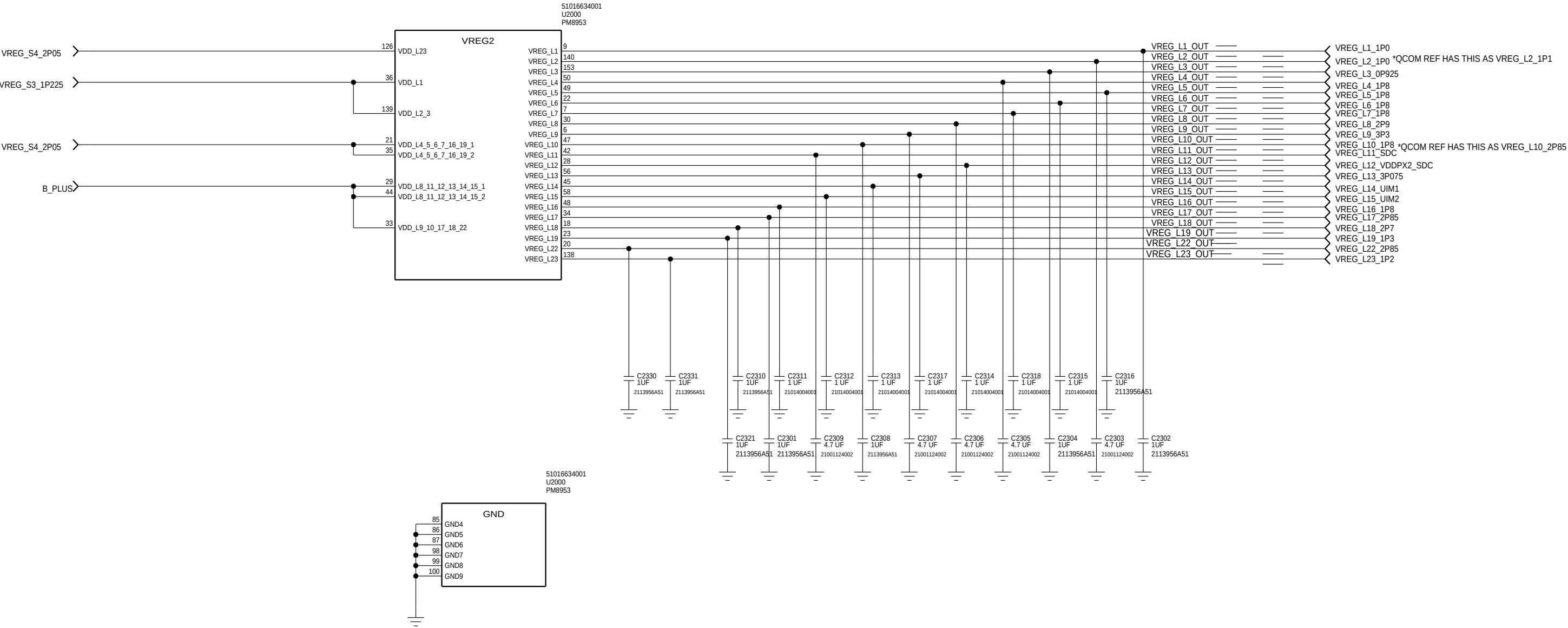


NOTE: PLACE INPUT CAPS CLOSE TO RESPECTIVE VIN PINS
EACH CAP GND SHOULD BE STAR-ROUTED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Tue Nov 15 13:46:05 2016			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B houdini_a.cpm	REV 01
		CAD FILE NAME:	PAGE 27 OF 64

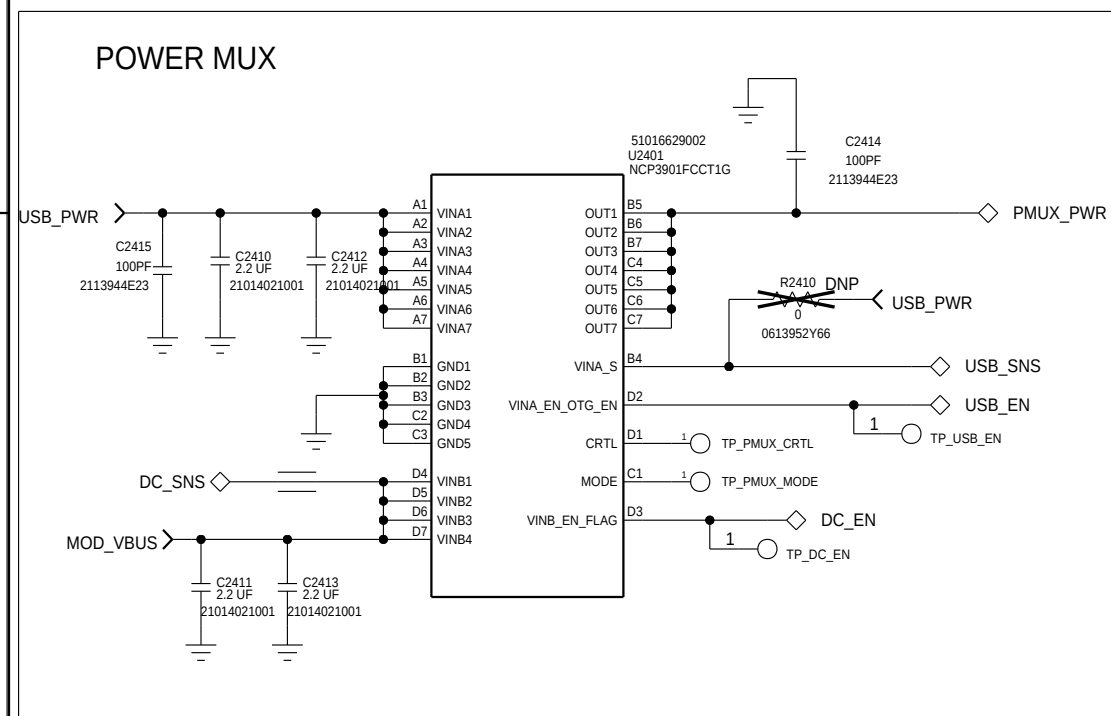
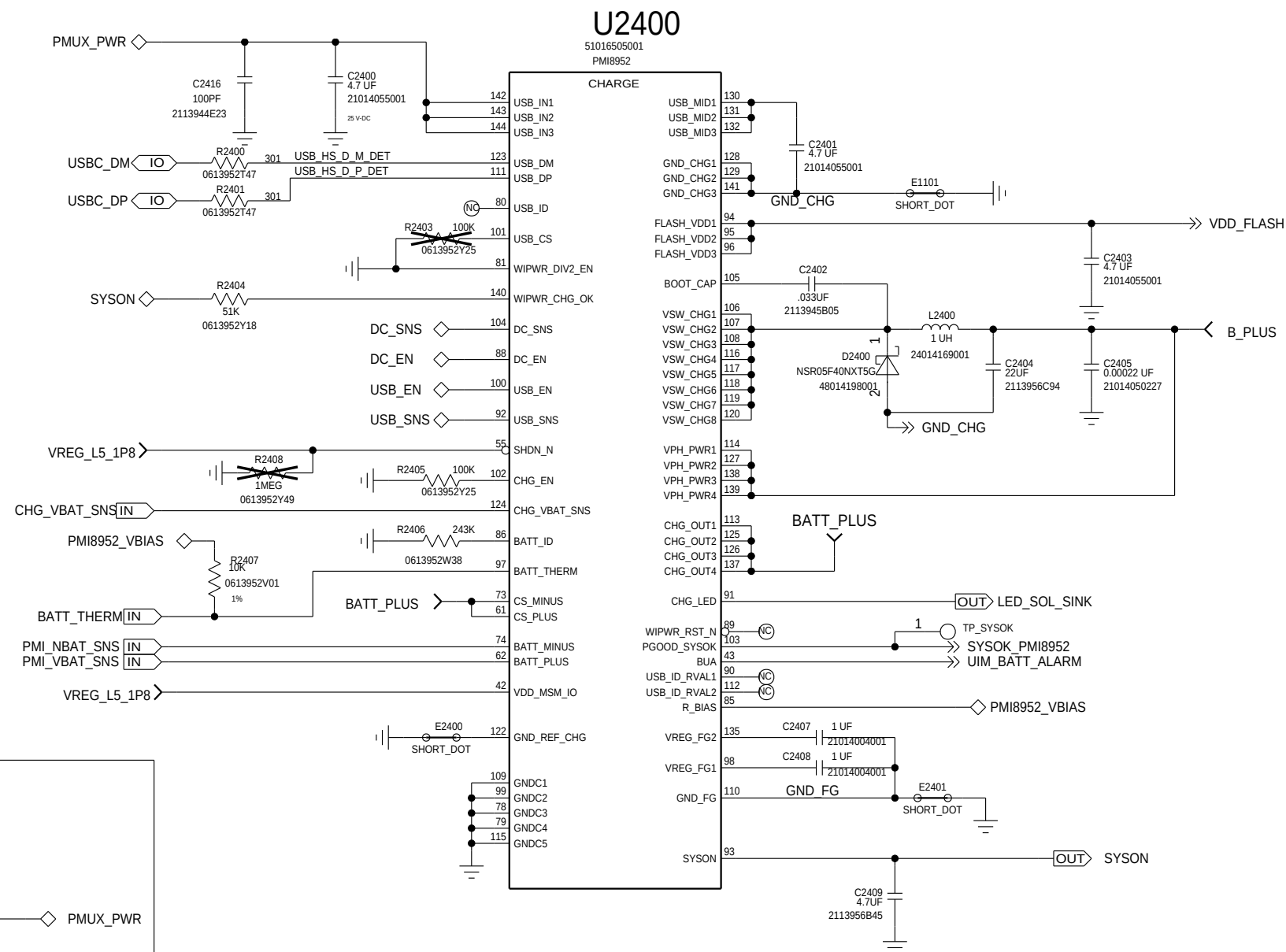
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REF: 2300-2399



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 28 OF 64	

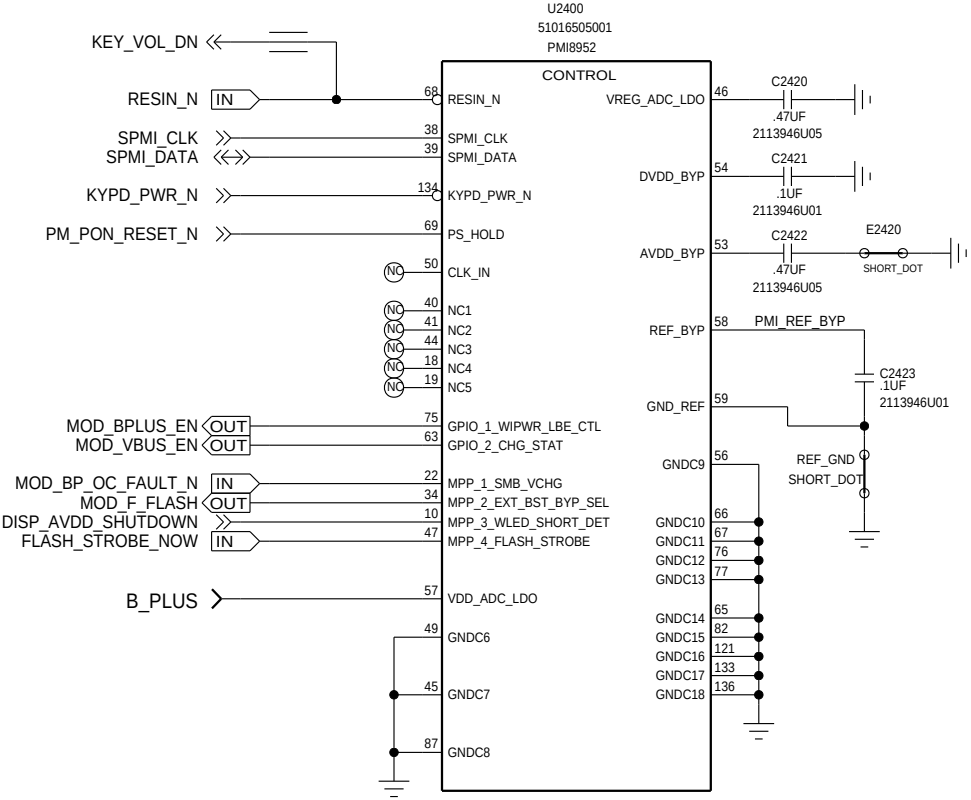
PMI8952: CHARGING
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APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI	BLOCK TITLE: ap_pmic			
LAST MODIFIED BY: MMI Thu Dec 22 15:29:34 2016				
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		CAD FILE NAME:		PAGE 29 OF 64

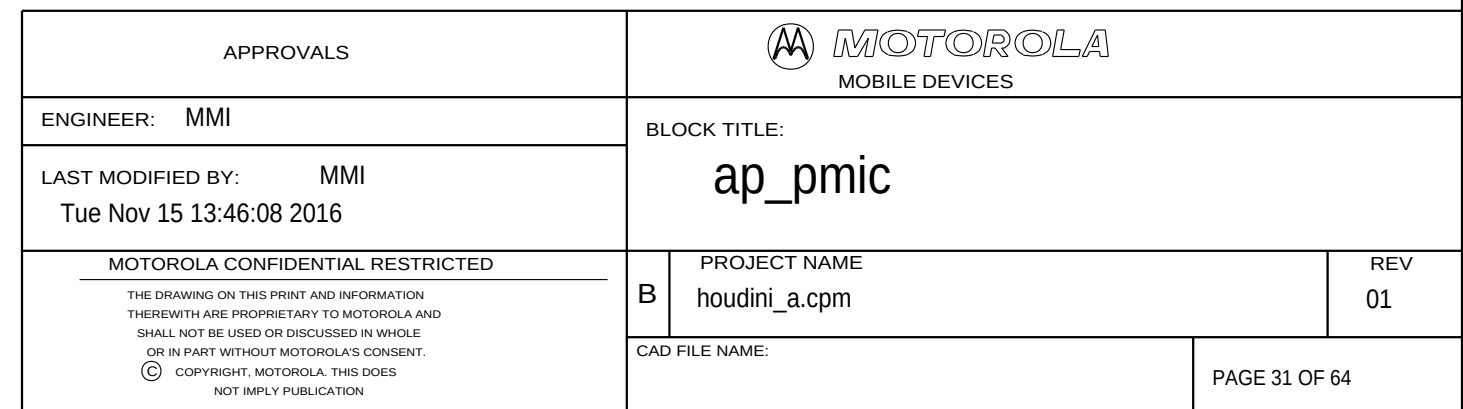
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REF: 2420-2439



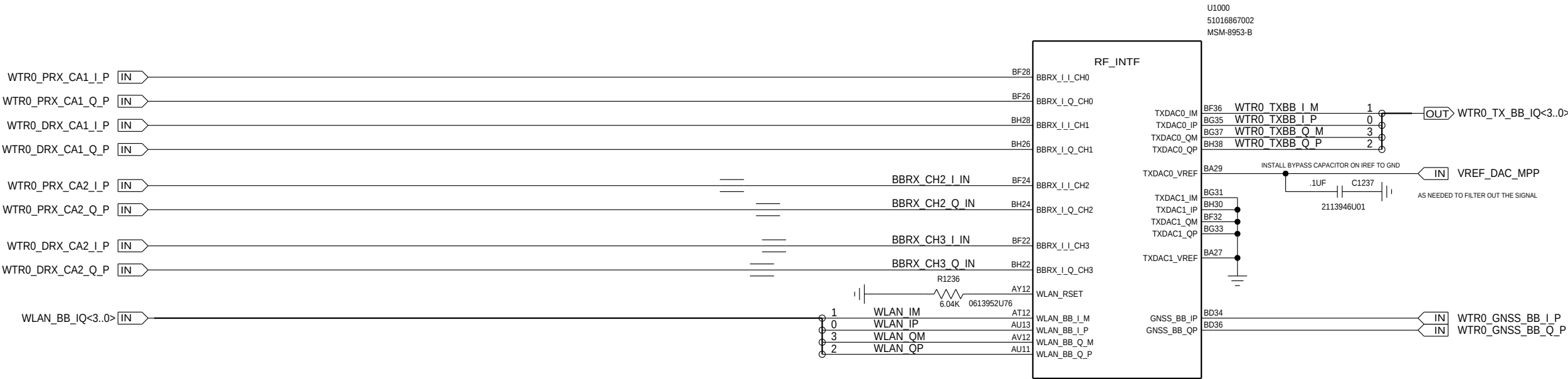
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		B houdini_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 30 OF 64	

REF: 2440-2459



MSM: RF INTERFACE

REF 1200-1299



NEED TO APPLY LC FILTER FOR REDUCE B40 RX SPURS(QC 80-NP237-56)

LC FILTER COULD BE PLACED ANYWHERE BETWEEN WTR BBRX_IQ AND MSM BBRX_IQ PINS

BUT THE GND FOR SHUNT C SHOULD NOT BE SHARED WITH EITHER WTR OR MSM GND ON ANY LAYER.

A DIRECT VIA FROM SHUNT C TO MAIN GND IS RECOMMENDED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: block_mdm9x25m_pm8019	
LAST MODIFIED BY: FAN021 Tue Nov 15 13:45:58 2016			
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		CAD FILE NAME:	
		PAGE 32 OF 64	

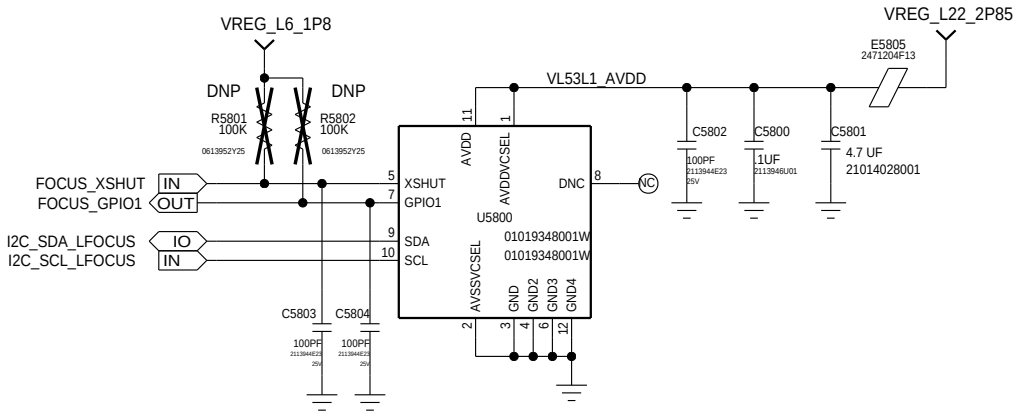
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REF:5800-5899

REAR LASER FOCUS

REF:6190-6199

I2C ADDR:0X29

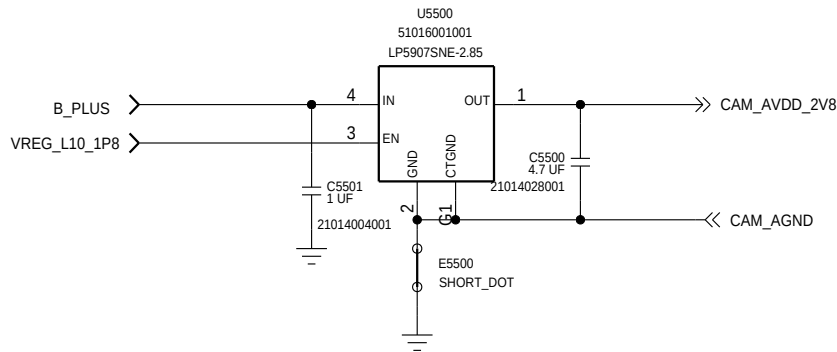


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
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		B houdini_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 33 OF 64	

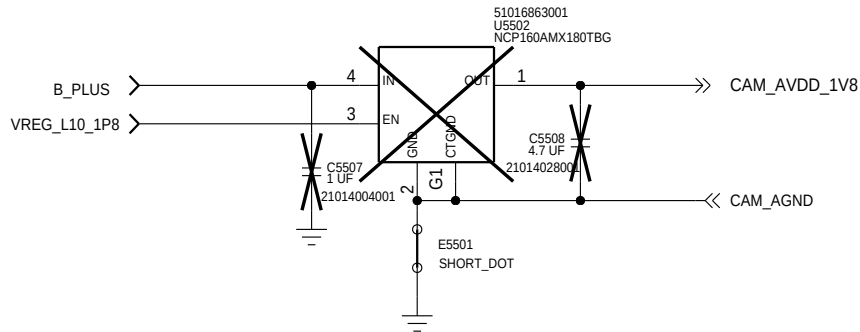
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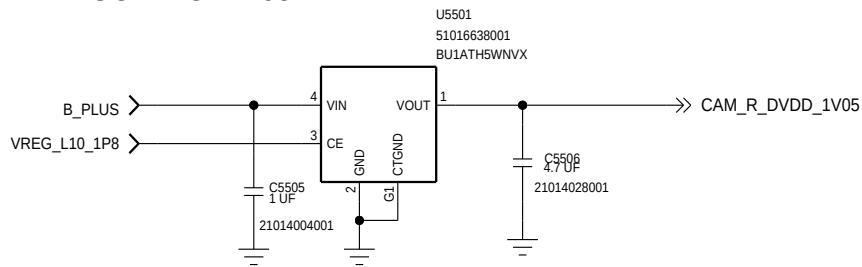
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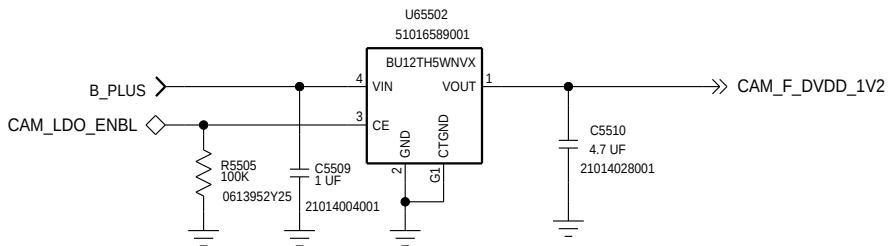
ANALOG REGULATOR 1.8V



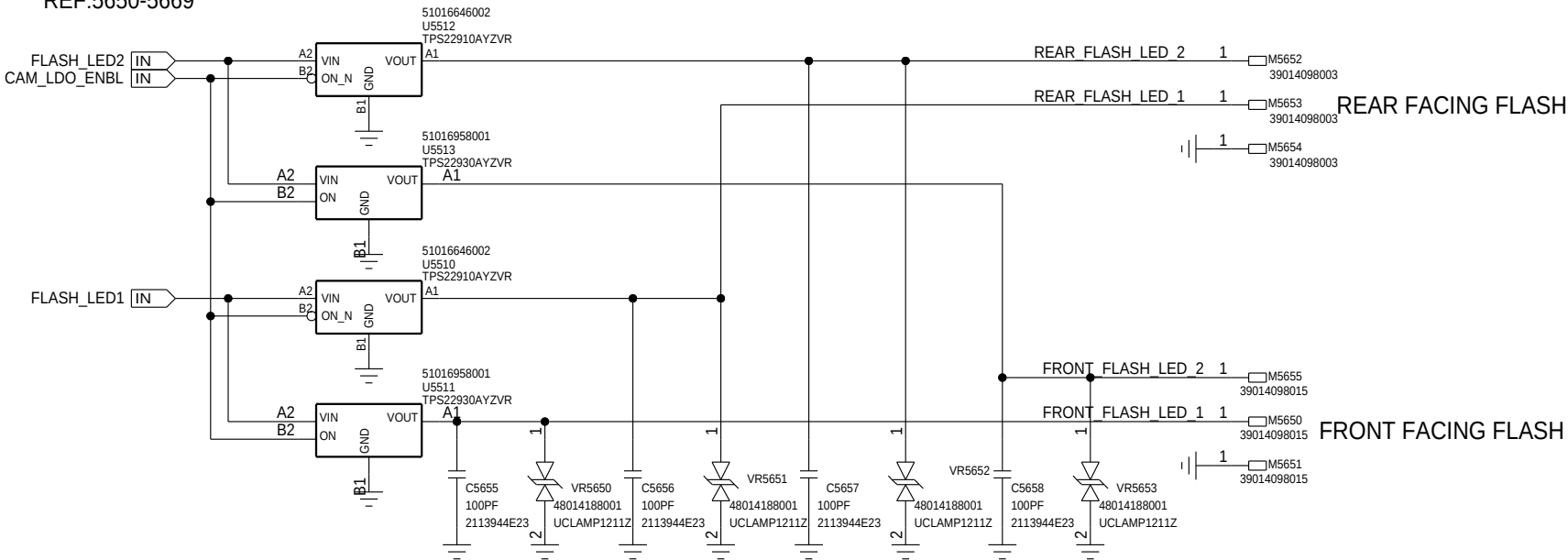
DIGITAL REGULATOR 1.05V



CAMERA CORE - 1.2V



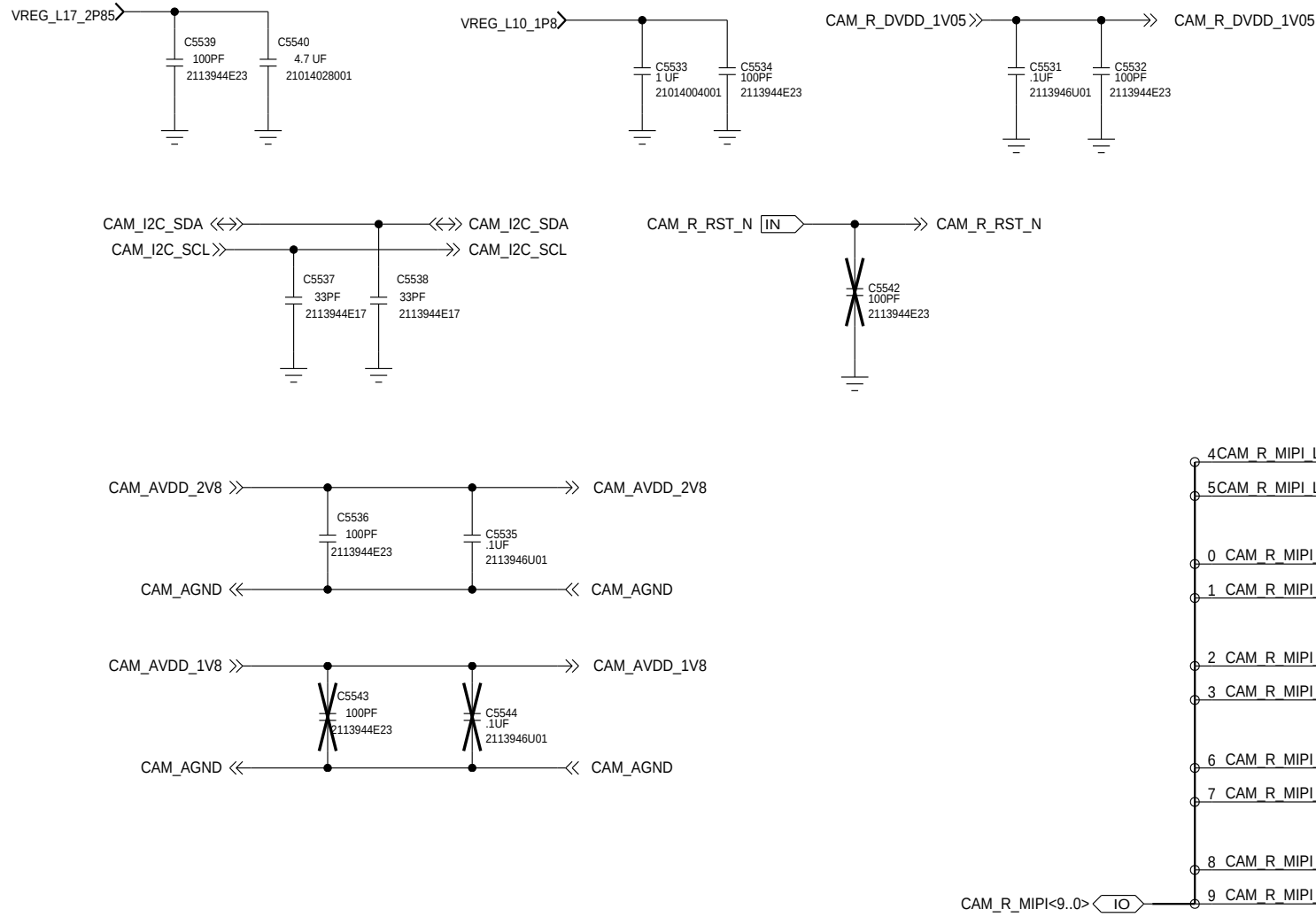
LED MUX FOR REAR VS. FRONT FLASH
REF:5650-5669



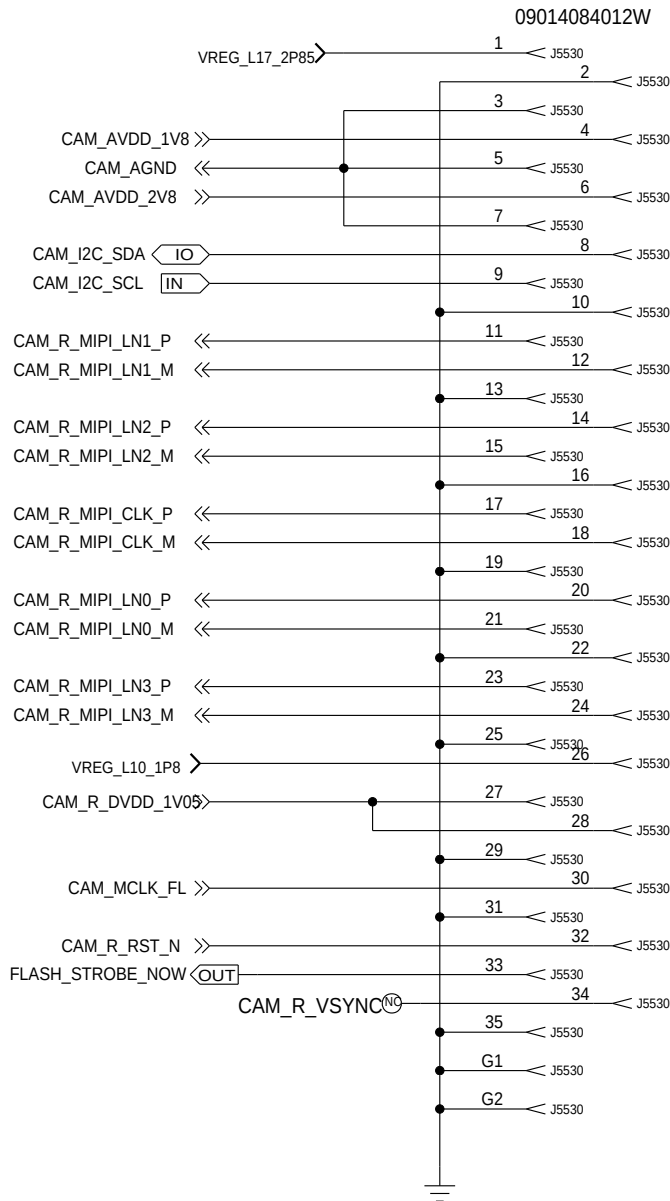
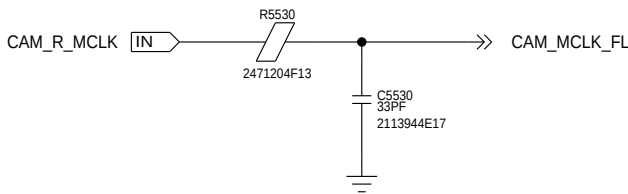
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		PAGE 34 OF 64	

CAM: REAR

REF:5530-5549



EMI/ESD FILTER

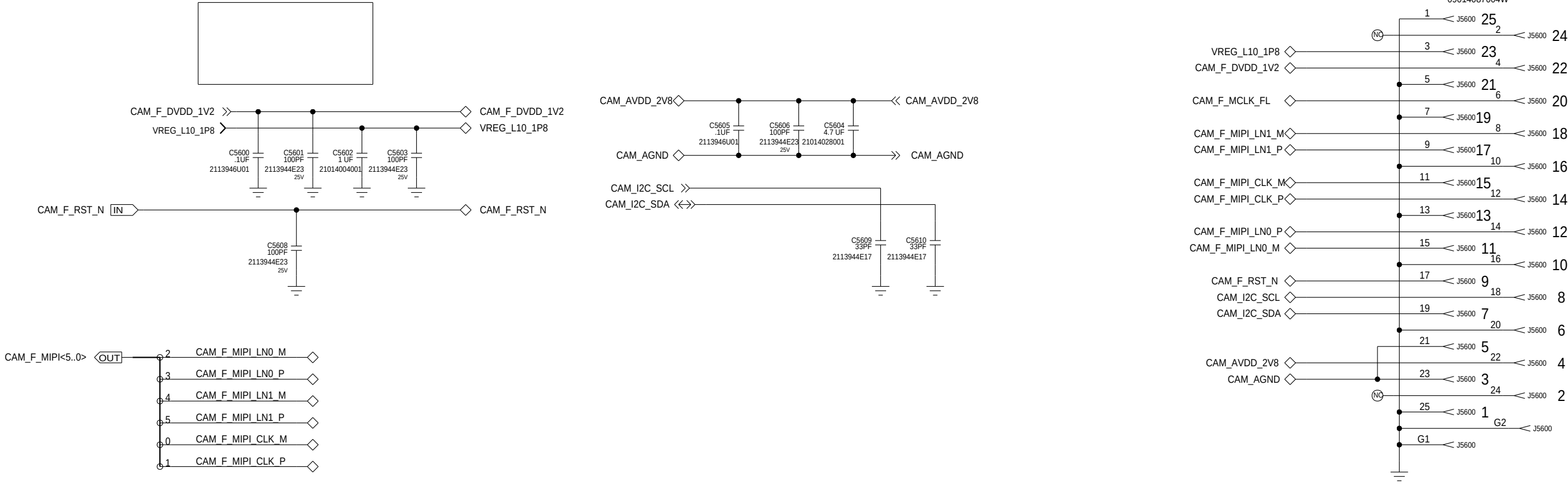


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LAST MODIFIED BY: AZIZ NOORI Tue Nov 15 13:45:54 2016		PROJECT NAME houdini_a.cpm	
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CAD FILE NAME:		PAGE 35 OF 64	

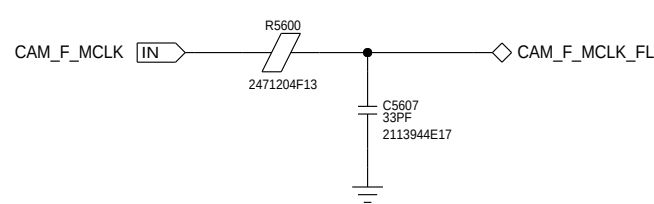
CAM: FRONT

REF 5600-5620

FF CAMERA CONNECTOR FRONT CAM I2C ADDR:

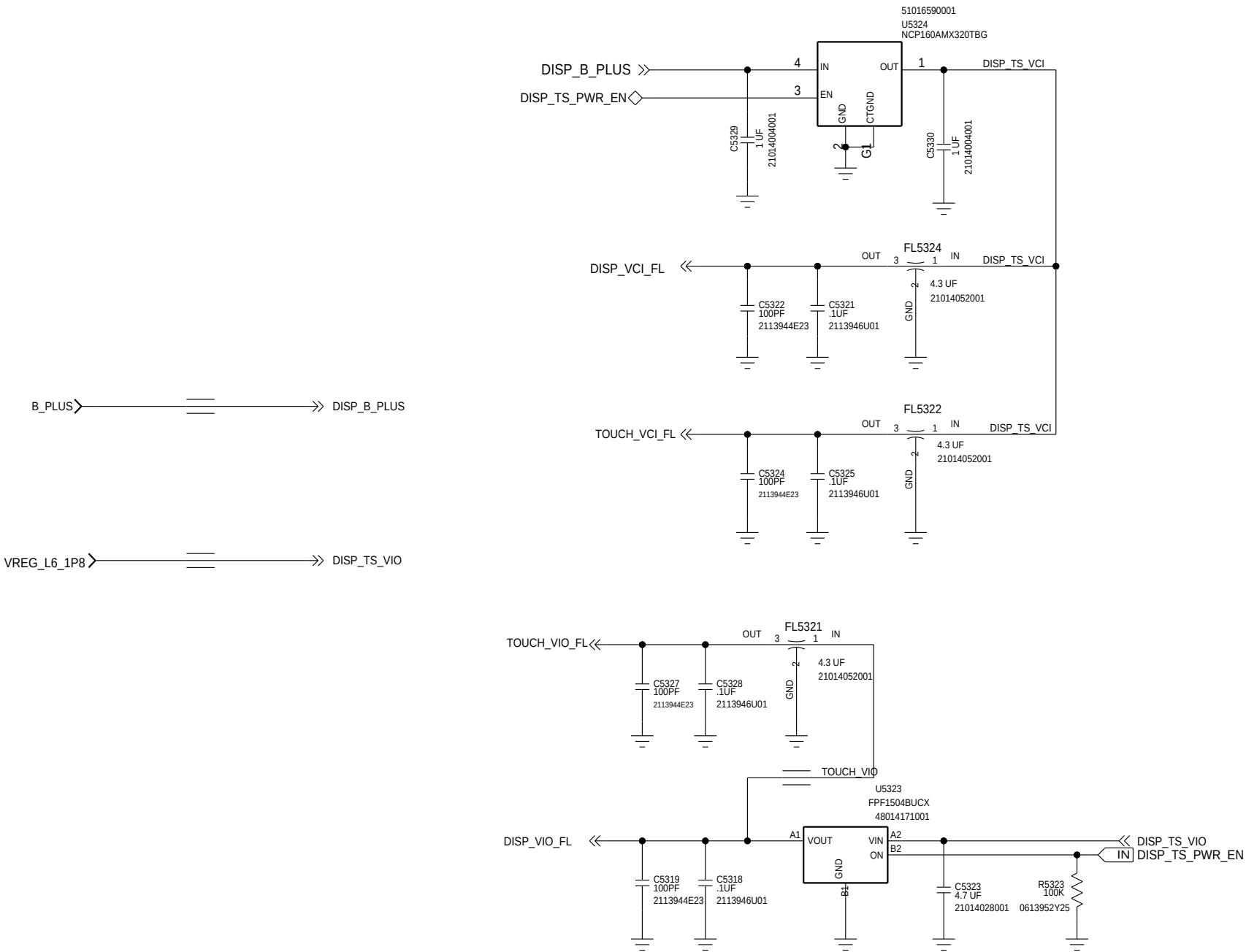


EMI/ESD FILTER



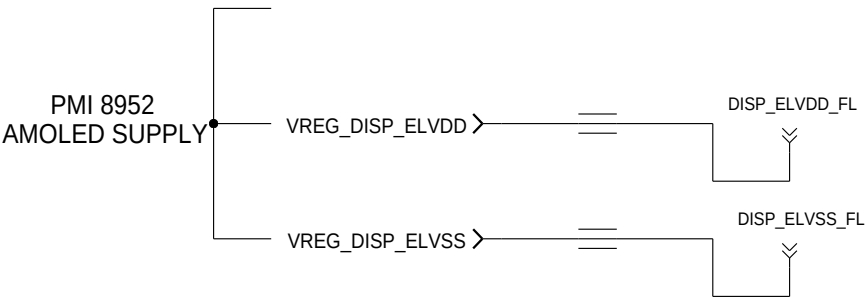
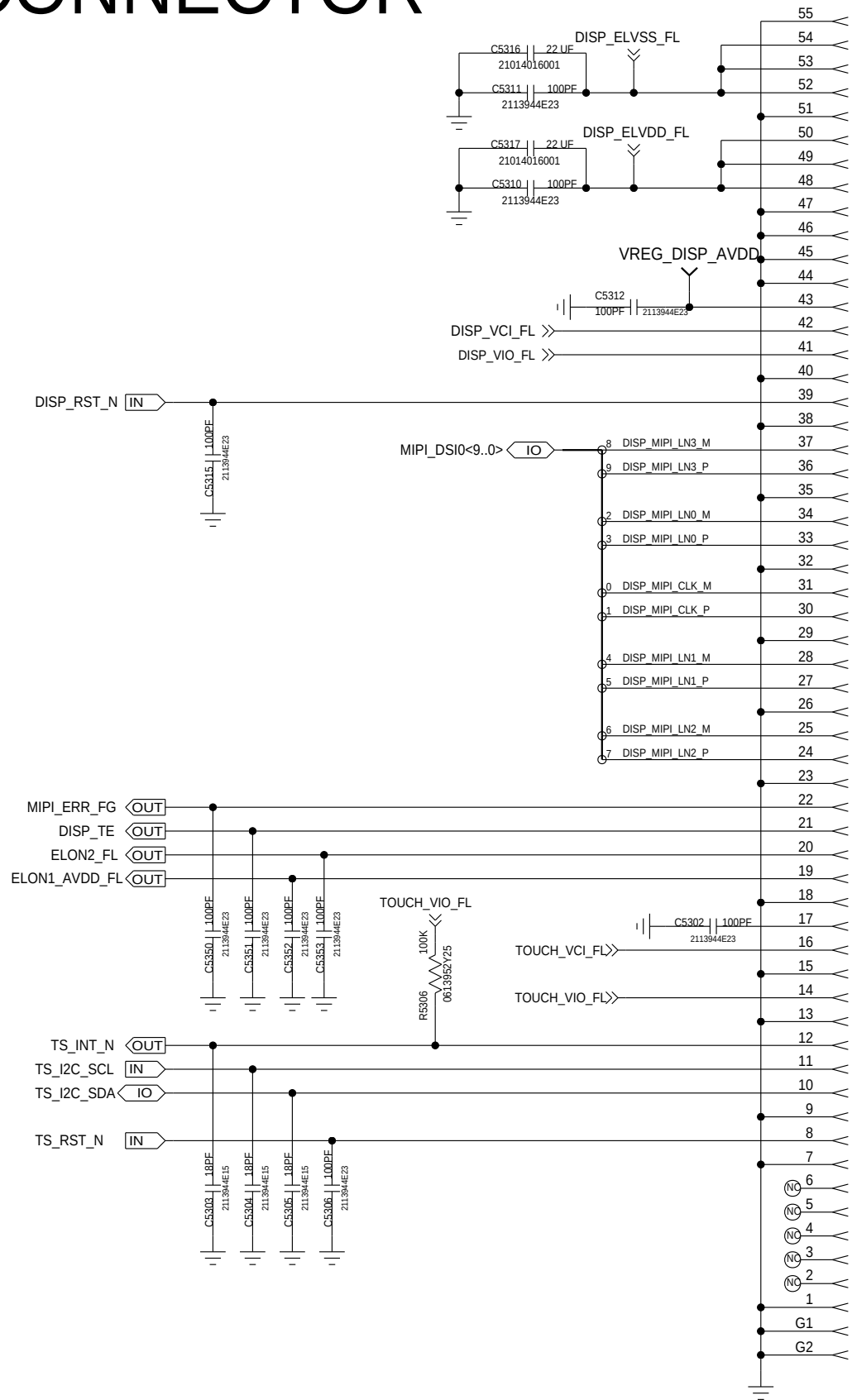
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LAST MODIFIED BY: NATE CONNELL Tue Nov 15 13:45:55 2016		multimedia	
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DISP: POWER



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: multimedia		
LAST MODIFIED BY: MMI Tue Nov 15 13:45:55 2016				
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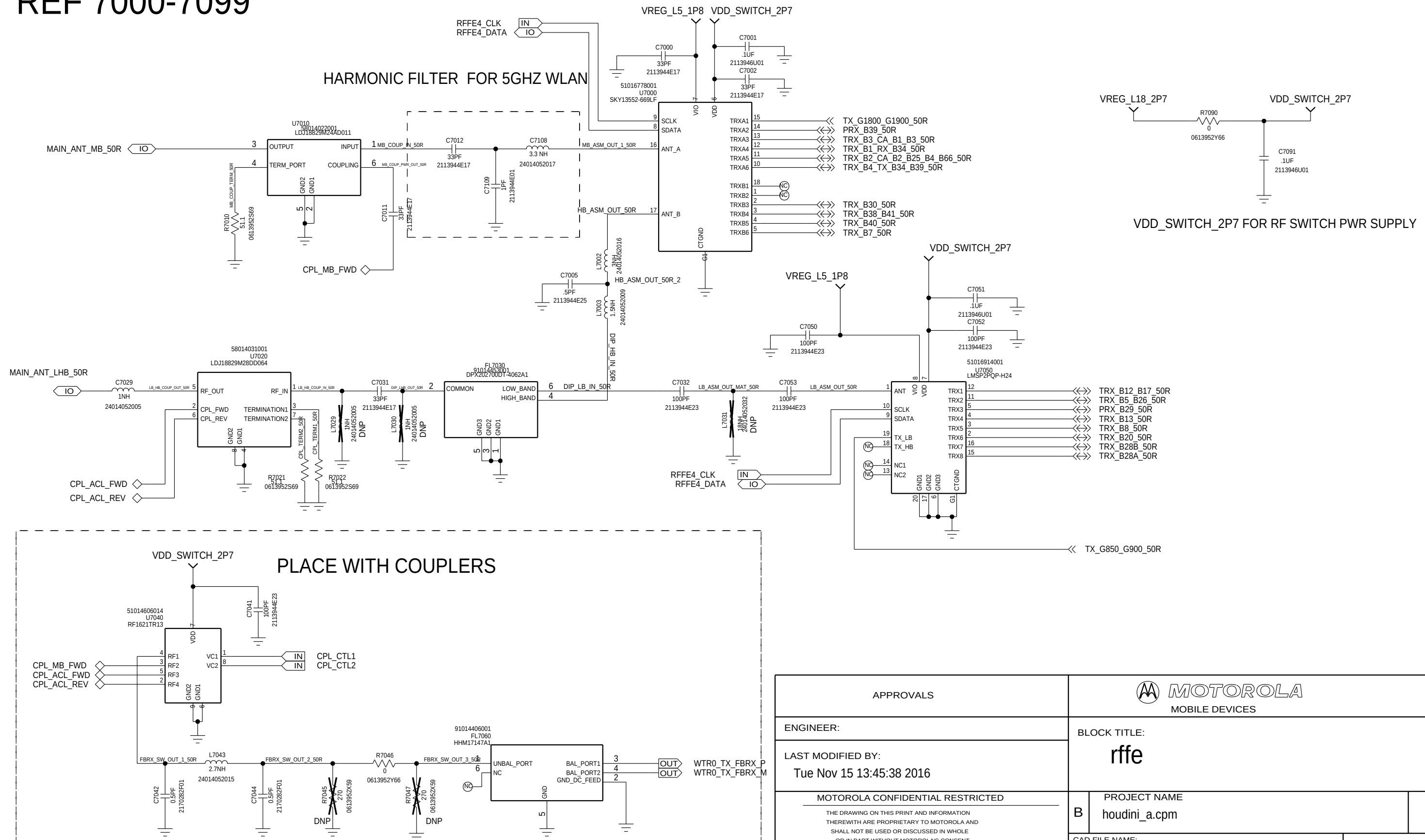
DISP: CONNECTOR



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		B houdini_a.cpm	
		REV 01	
		PAGE 38 OF 64	

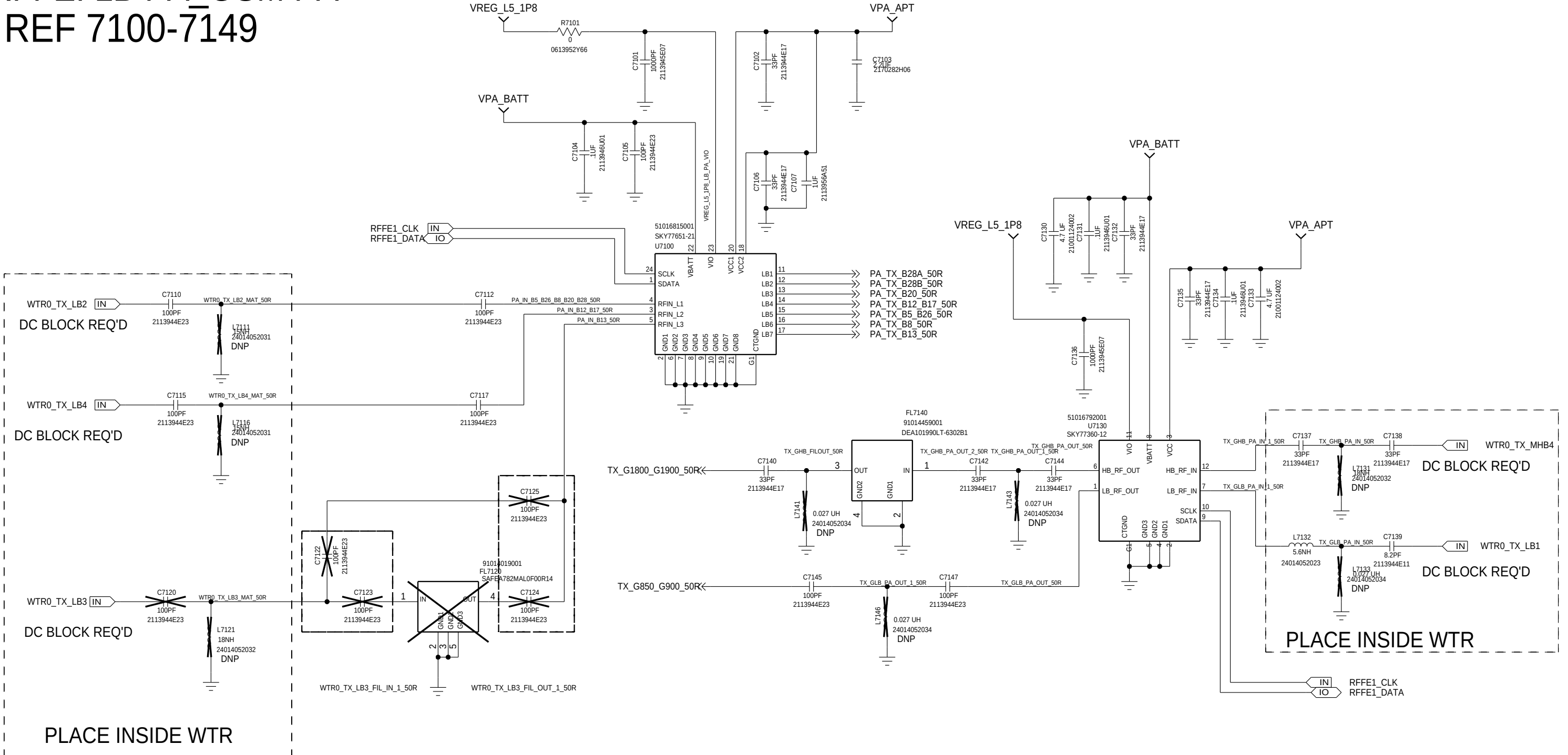
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MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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RFFE: LB PA_GSM PA
REF 7100-7149



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe		
LAST MODIFIED BY: Thu Dec 22 09:39:49 2016			
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		CAD FILE NAME:	PAGE 46 OF 64

D



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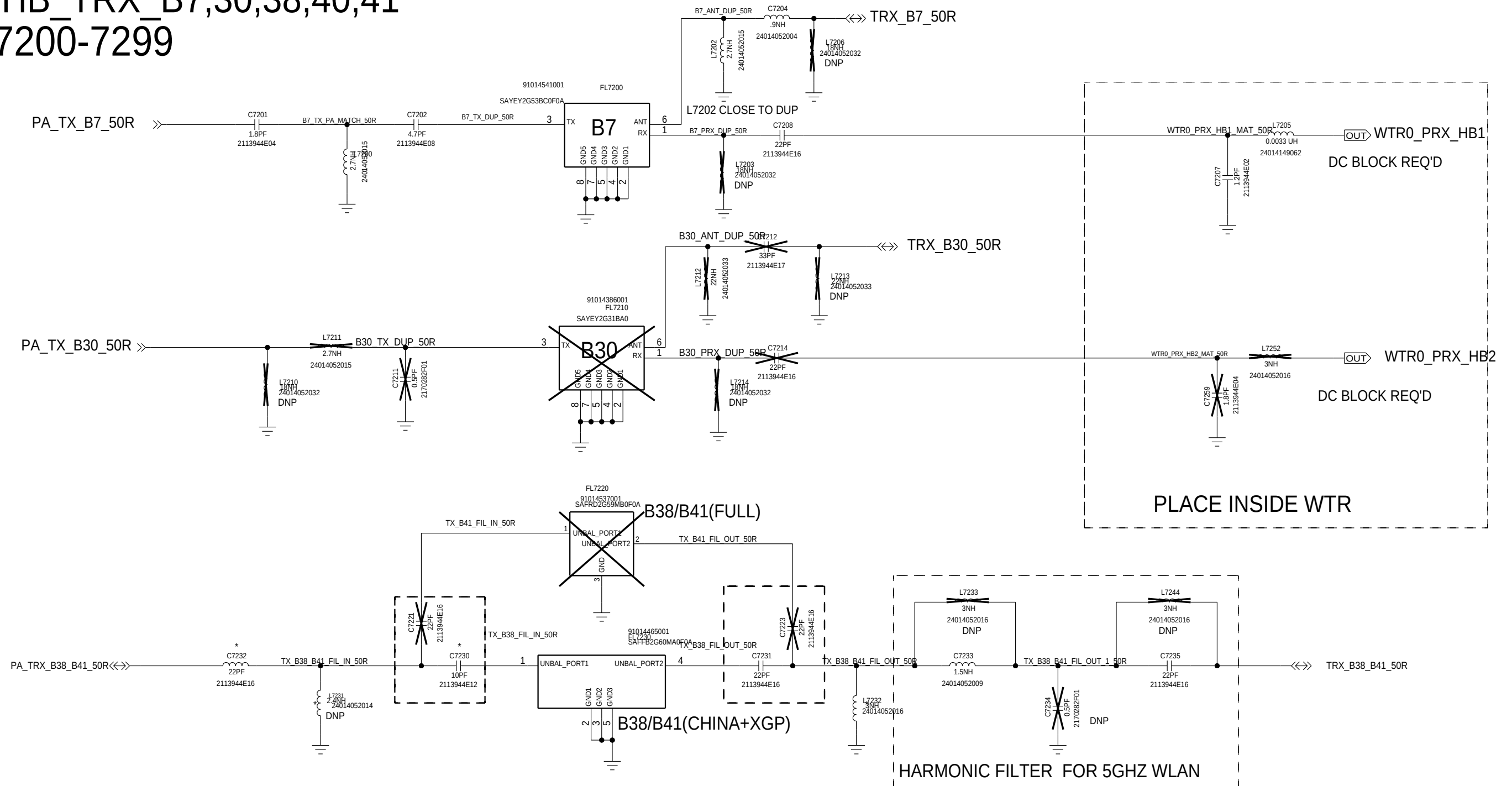
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RFFE: HB_TRX_B7,30,38,40,41
REF 7200-7299



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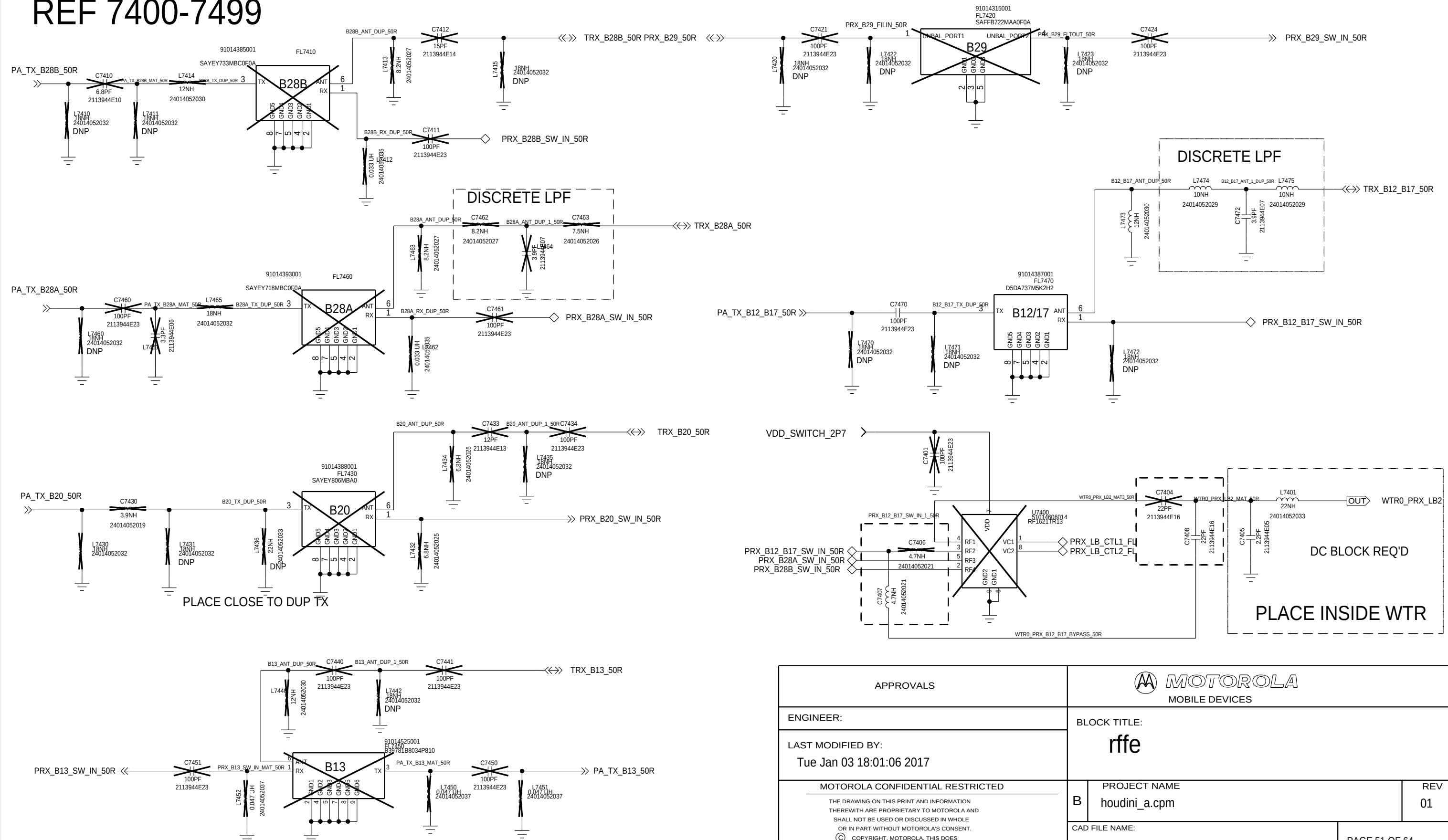
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LAST MODIFIED BY: Tue Jan 03 18:01:06 2017			
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ENGINEER:

LAST MODIFIED BY:

Wed Nov 23 10:22:25 2016

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MOTOROLA
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BLOCK TITLE:

rffe

	PROJECT NAME
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B	houdini_a.cpm
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CAD FILE NAME:

EV
1

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D



DC BLOCK REQ'D

DC BLOCK REQ'D

80-NH379-45

DC BLOCK REQ'D

DC BLOCK REQ'D

PLACE INSIDE WTR



APPROVALS

ENGINEER:

LAST MODIFIED BY:

Fri Dec 02 11:48:33 2016

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BLOCK TITLE:

rffe

	PROJECT NAME
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B	houdini_a.cpm
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CAD FILE NAME:

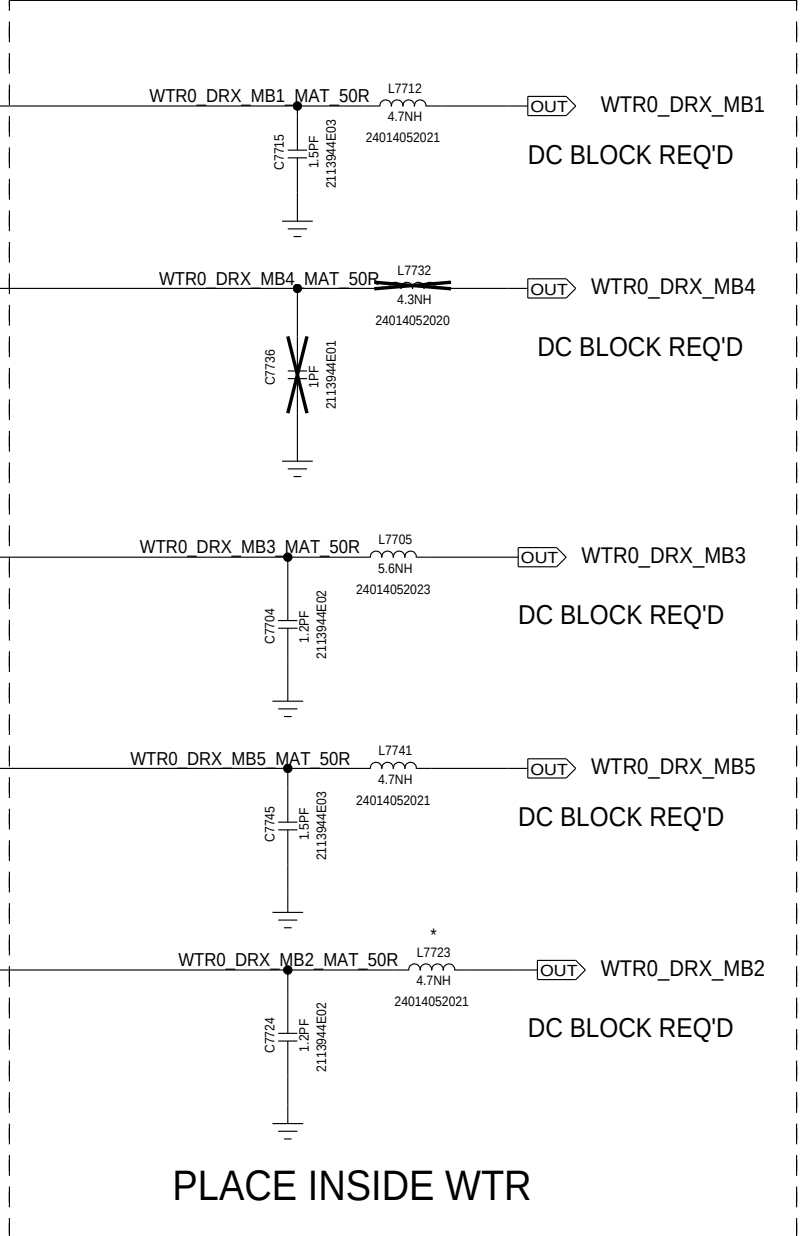
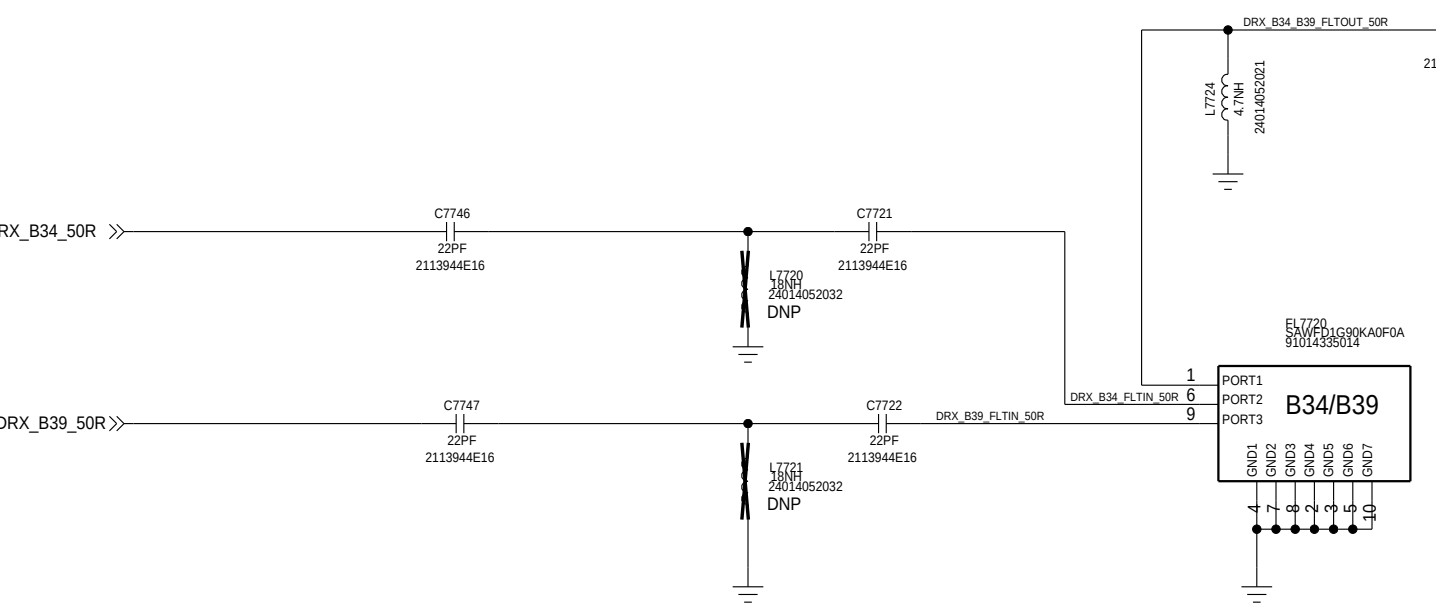
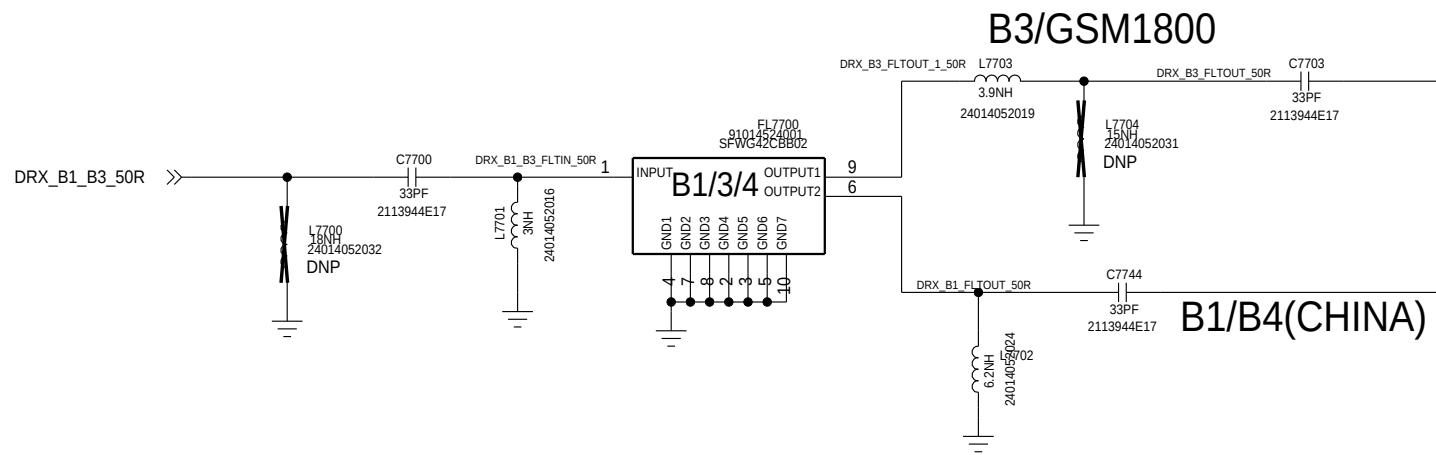
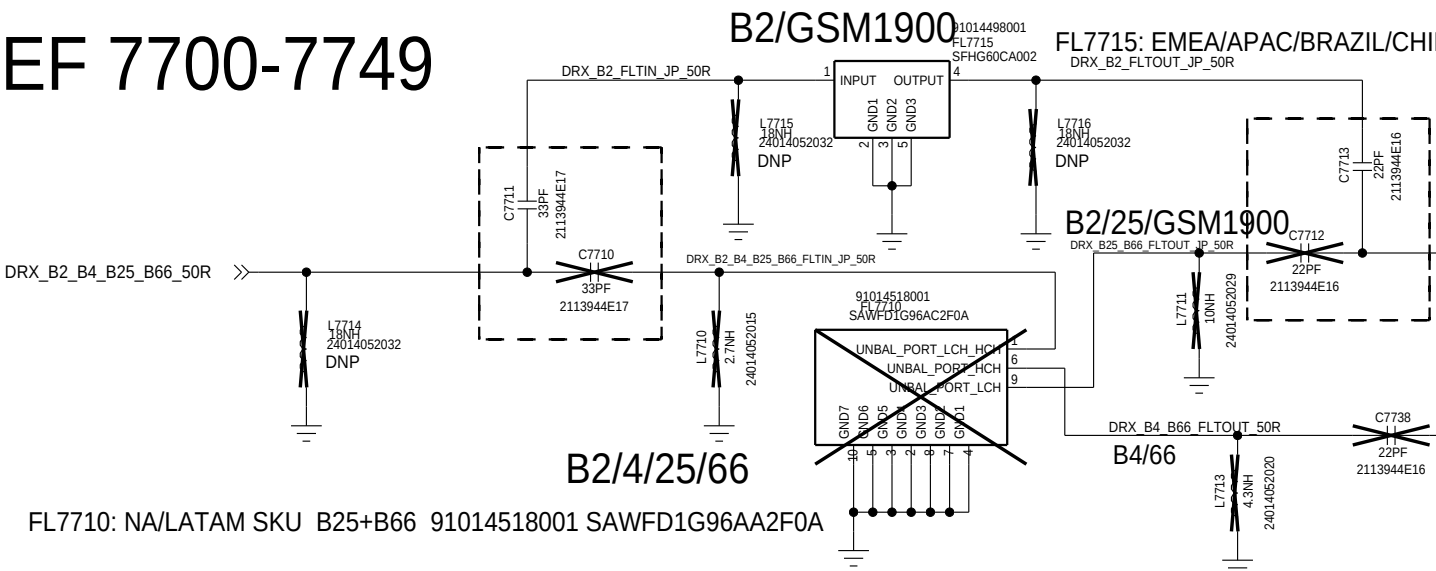
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RFFE: MB_DRX_B1,2,3,4,,25,34,39,66

REF 7700-7749



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Tue Nov 22 17:48:25 2016		PROJECT NAME houdini_a.cpm	
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			PAGE 54 OF 64

D



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B

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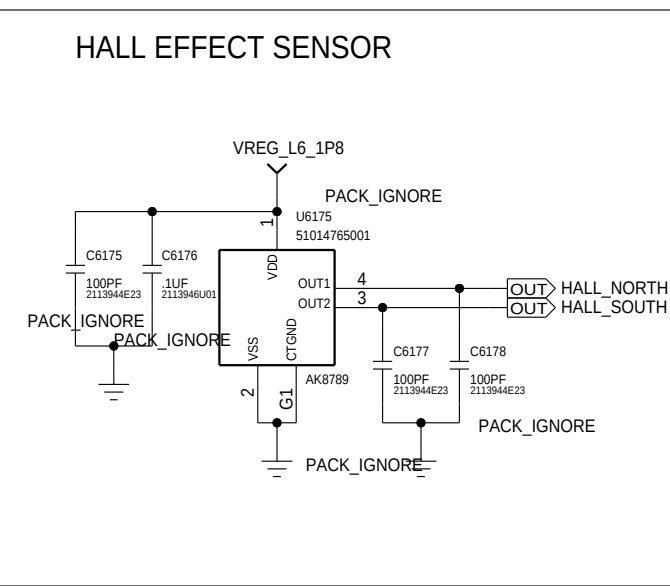
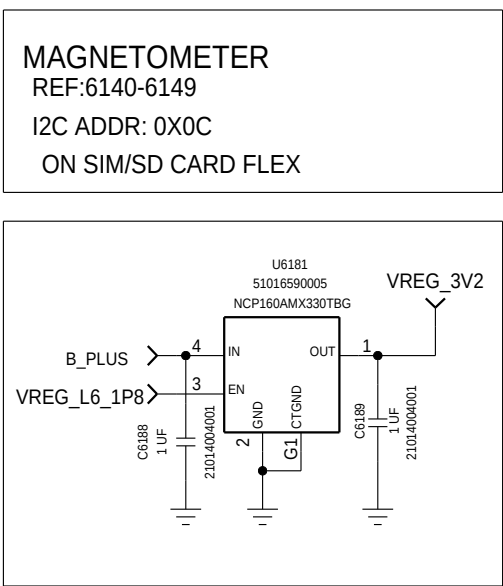
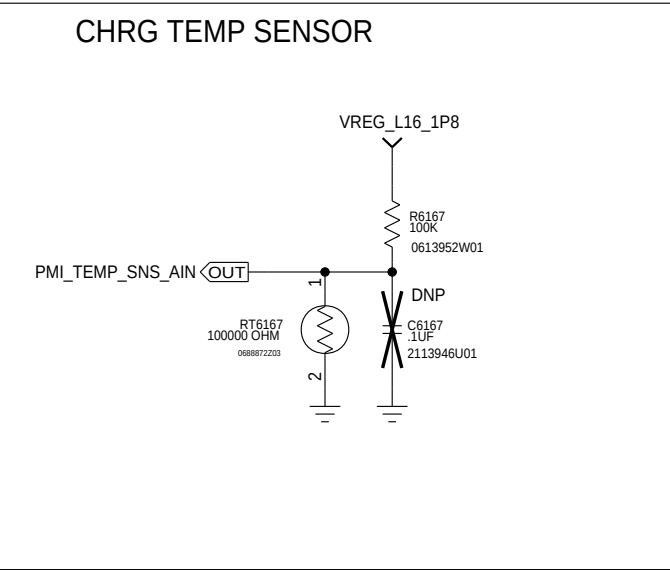
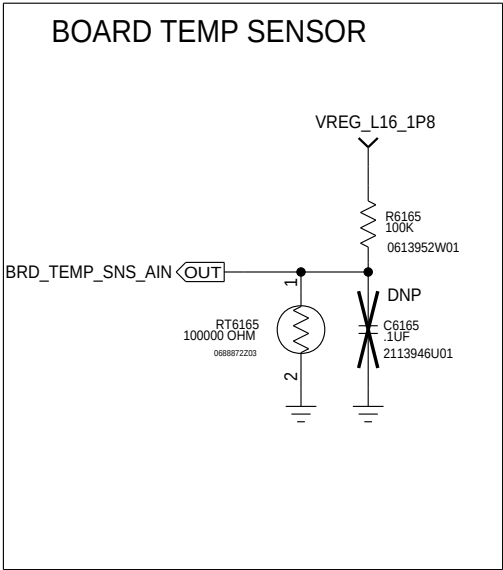
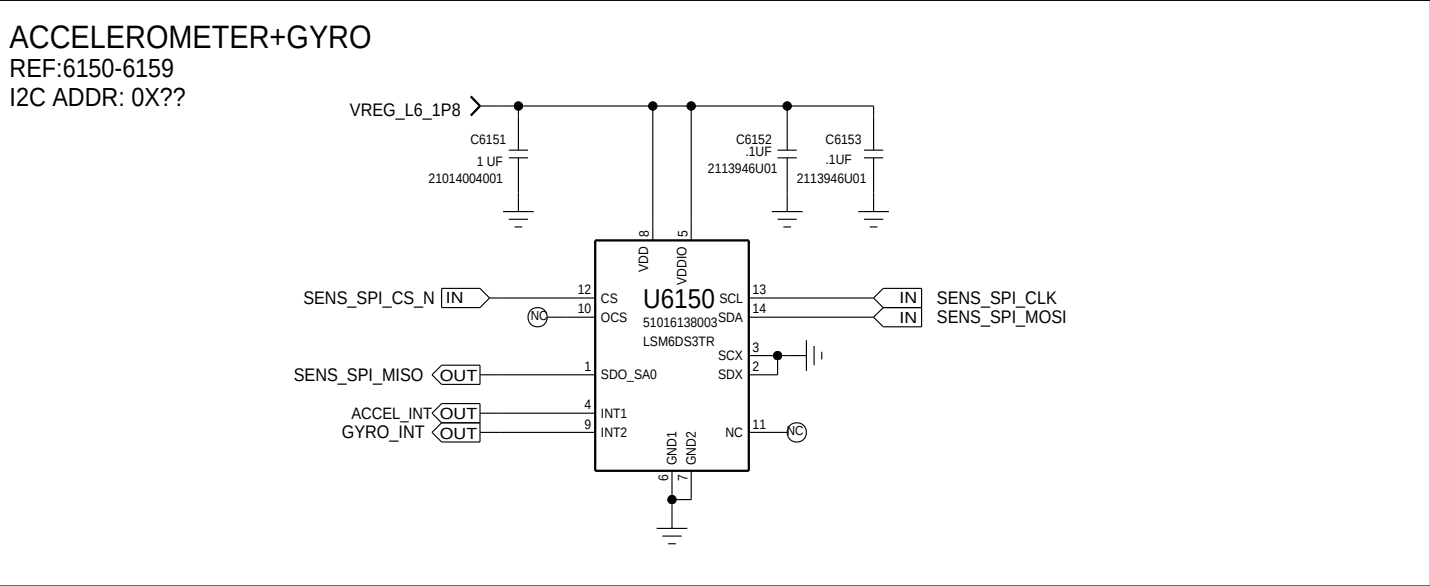
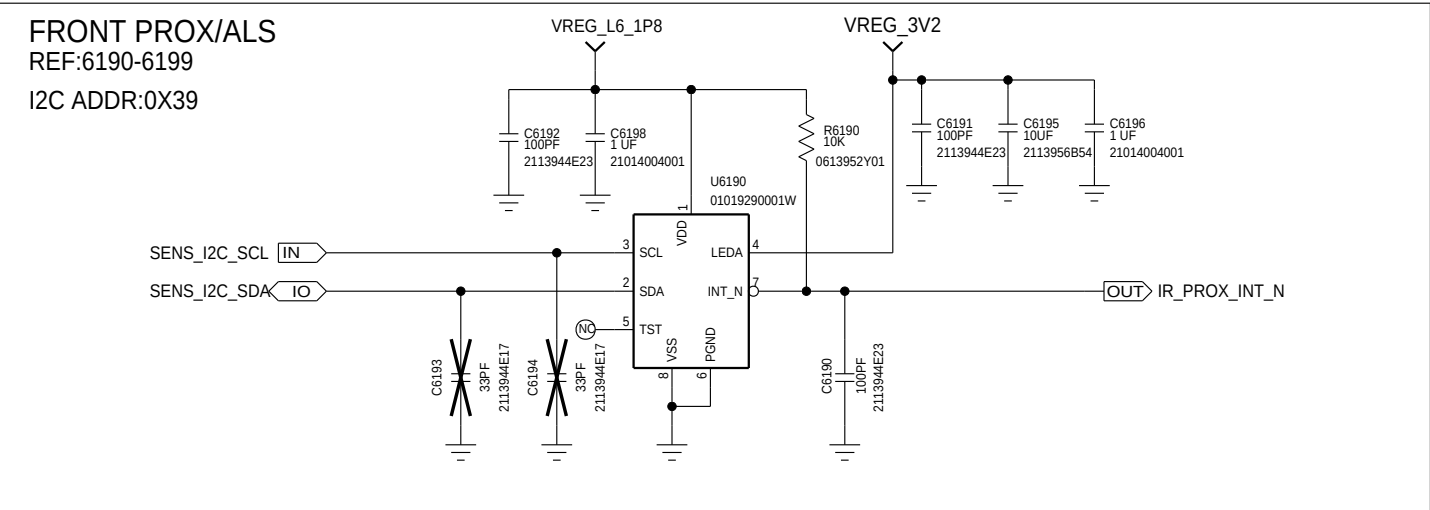


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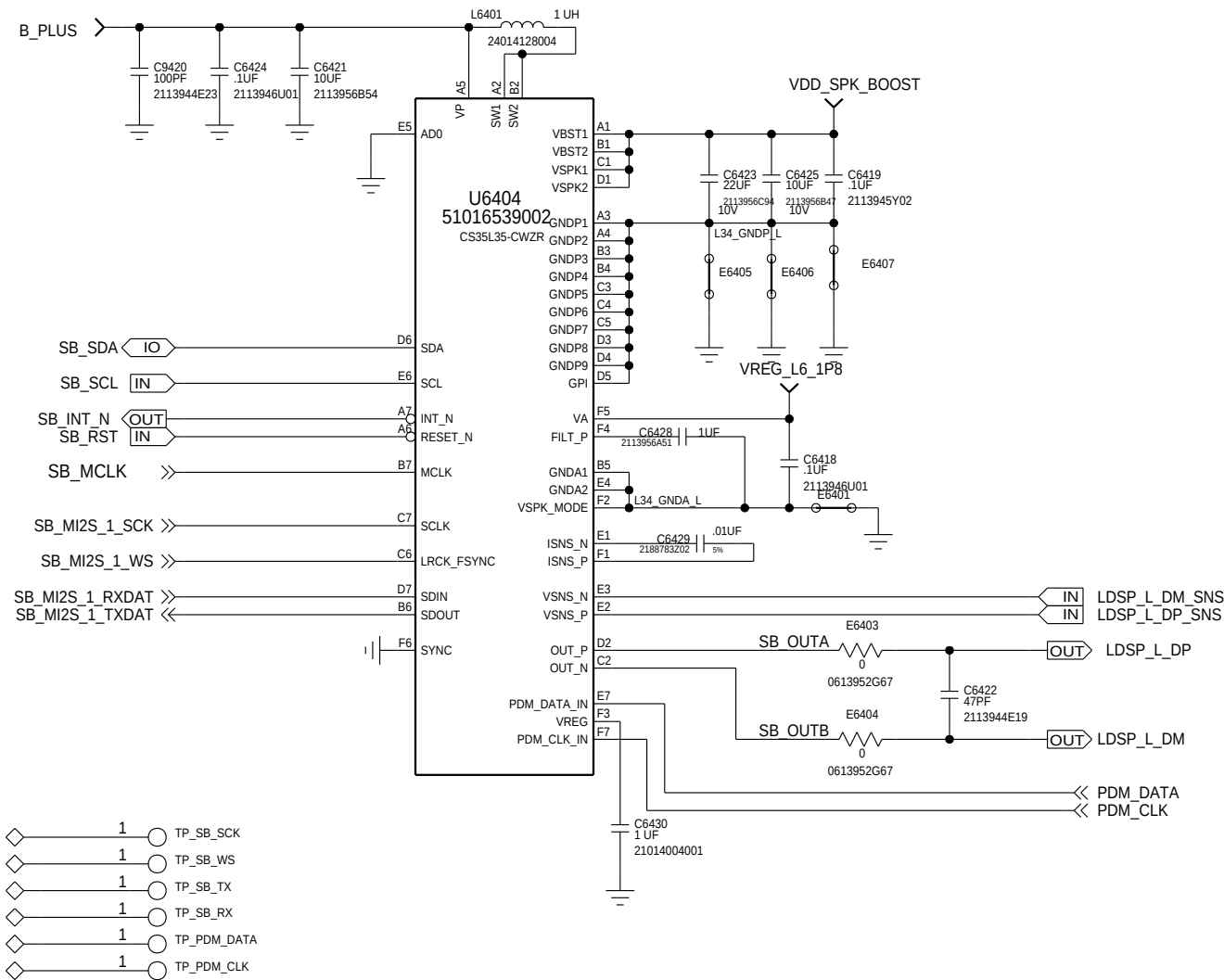


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		REV 01 PAGE 58 OF 64	

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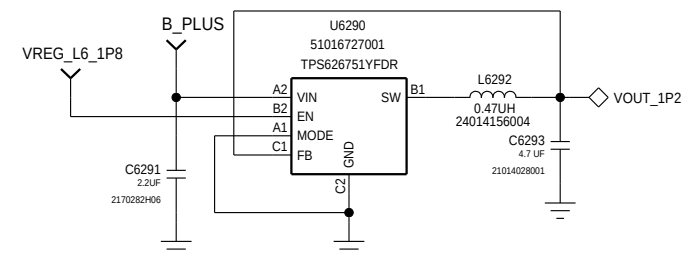
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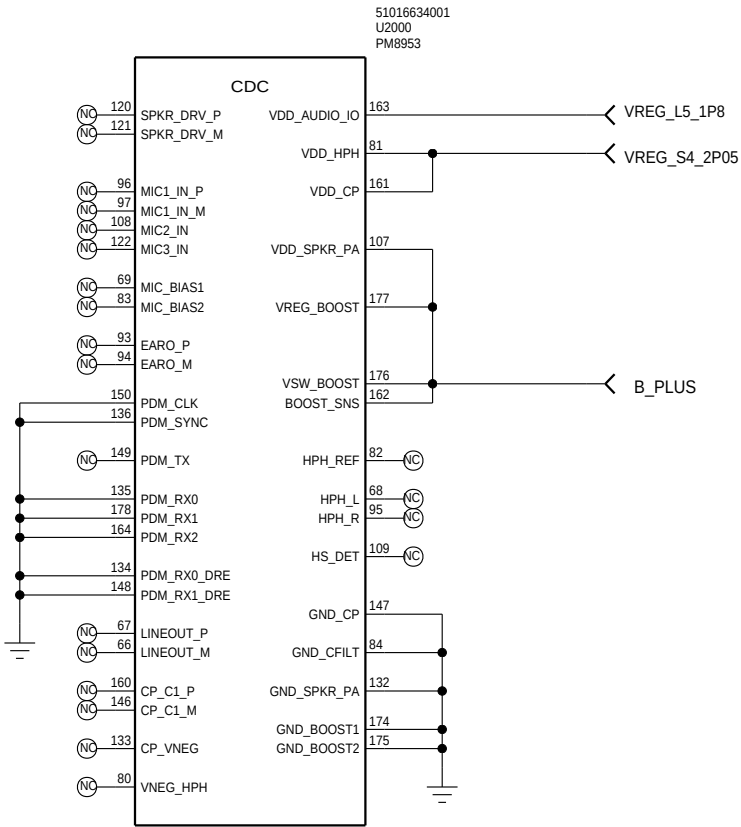
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		CAD FILE NAME:	PAGE 60 OF 64

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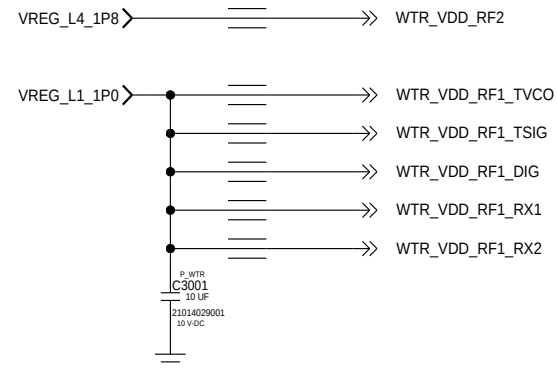
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		PAGE 61 OF 64	

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APPROVALS



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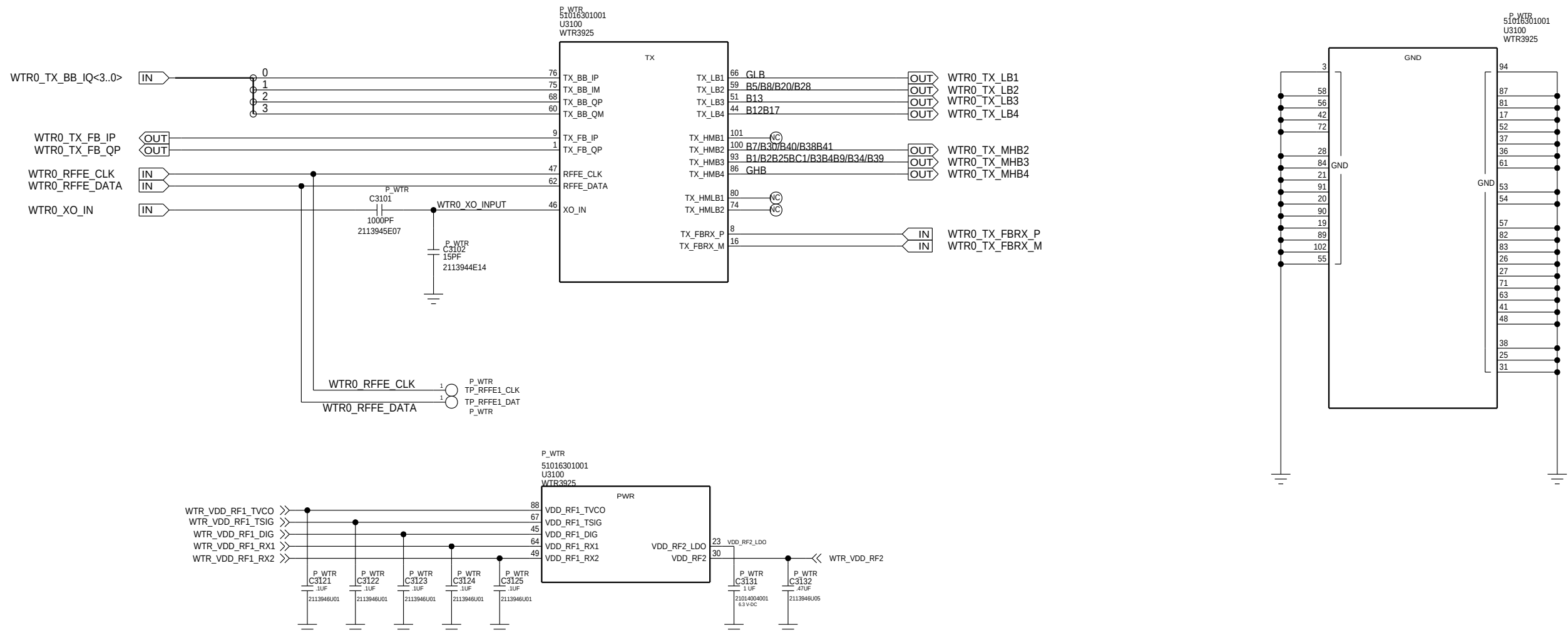
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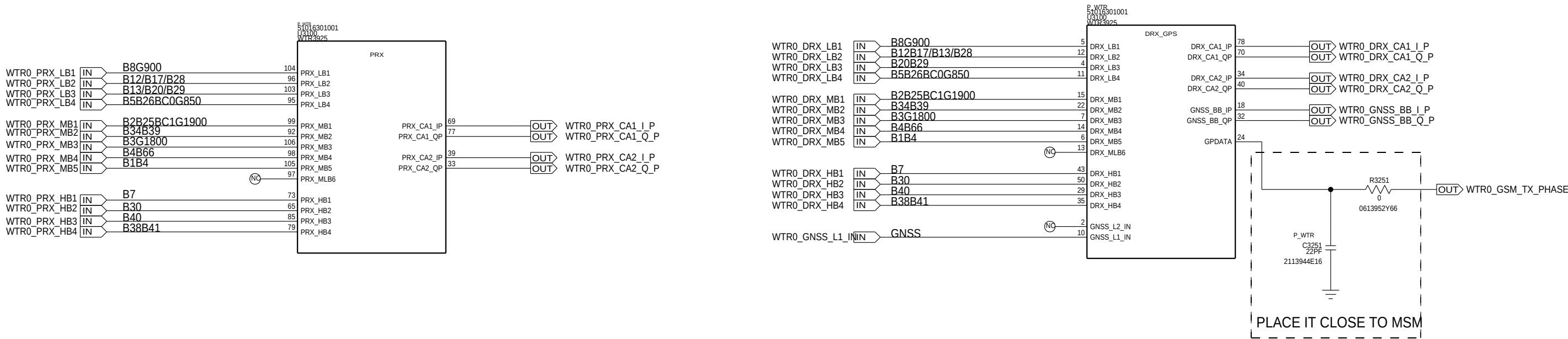
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WTR3925: PRX/DRX/GPS

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